

Consensus statement: An evidence-based review of exercise, rehabilitation, rest, and return to activity protocols for the treatment of concussion and mild traumatic brain injury

Arthur J. De Luigi DO, MHSA¹  | Kathleen R. Bell MD²  |
 Joe P. Bramhall MD³ | Meeryo Choe MD⁴ | Katherine Dec MD⁵ |
 Jonathan T. Finnoff DO^{6,7} | Mark Halstead MD⁸ | Stanley A. Herring MD⁹ |
 Jason Matuszak MD¹⁰ | P. B. Raksin MD¹¹ | Jennifer Swanson BS, MEd¹² |
 Carolyn Millett BS¹³

¹Department Chair of Physical Medicine & Rehabilitation, Medical Director of Sports Medicine, Mayo Clinic, Scottsdale, Arizona, USA

²Department of Physical Medicine and Rehabilitation, University of Texas Southwestern Medical Center, Dallas, Texas, USA

³Texas A&M University, College Station, Texas, USA

⁴UCLA Health, Los Angeles, California, USA

⁵Virginia Commonwealth University, Richmond, Virginia, USA

⁶United States Olympic and Paralympic Committee, Colorado Springs, Colorado, USA

⁷Department of Physical Medicine and Rehabilitation, University of Colorado, Denver, Colorado, USA

⁸Washington University Sports Medicine, St Louis, Missouri, USA

⁹Department of Rehabilitation Medicine, University of Washington, Seattle, Washington, USA

¹⁰Sports Medicine, Excelsior Orthopaedics, Buffalo, New York, USA

¹¹John H. Stroger Jr Hospital of Cook County (formerly Cook County Hospital), Rush University Medical Center, Chicago, Illinois, USA

¹²Swanson Medical Writing, Masonville, Colorado, USA

¹³AAPM&R, Rosemont, Illinois, USA

Correspondence

Arthur J. De Luigi, Department Chair of Physical Medicine & Rehabilitation, Medical Director of Sports Medicine, Mayo Clinic, Scottsdale, AZ, USA.
 Email: ajweege@yahoo.com

Abstract

Background: Early diagnosis and appropriate management of concussion/mild traumatic brain injury (mTBI) is critical for preventing poor outcomes and minimizing health care burden. Current clinical guidelines for concussion management focus mostly on diagnosis and return to cognitive and physical activity but provide limited guidance on the use of specific therapeutic interventions.

Objective: To systematically review the available evidence on therapeutic interventions for concussion/mTBI and develop an evidence-based consensus statement on the use of these interventions in clinical practice.

Literature Survey: A systematic literature search was performed first in 2018 and 2019, and again in 2022, to identify relevant original research on these interventions. A total of 6303 articles were retrieved through the systematic literature search and screened for inclusion. Eighty articles met inclusion criteria and were included in this review and consensus process.

Methodology: A multispecialty panel was convened to explore management of concussion/mTBI. Interventions evaluated included rest, exercise, rehabilitation, and return to activity (RTA) protocols. Studies were assessed for relevance and methodologic quality and were voted upon to develop an evidence-based consensus statement on the therapeutic appropriateness of these interventions for concussion/mTBI. A meta-analysis was not performed.

Synthesis: There was sufficient evidence to recommend exercise as an appropriate therapy for adolescents with acute concussion/mTBI. In other age groups and for other therapeutic modalities, although some studies demonstrated benefits for some of the interventions, mixed results and study limitations prevented the panel from drawing firm conclusions on the efficacy of those interventions. The panel found evidence of detrimental effects from strict rest and high-intensity physical activity.

Conclusions: The panel recommended exercise as an appropriate therapy for acute concussion in adolescents. The evidence on other therapeutic interventions for concussion/mTBI remains limited to small randomized controlled trials and observational studies of moderate to low quality. The panel found no strong evidence to support or recommend against the other evaluated interventions but found most interventions to be safe when used judiciously and in

consideration of individual patient needs. High-quality randomized studies with sufficient power are needed to evaluate the effects of rest, rehabilitation, and RTA protocols for the management of concussion/mTBI.

INTRODUCTION

Concussion encompasses a broad spectrum of clinical symptoms induced by biomechanical force to the head, face, or neck or by biomechanical forces to the body that are transmitted to the head.^{1–5} Concussions are considered equivalent to, or a subset of, mild traumatic brain injuries (mTBIs) that result in immediate and mostly transient disturbances of brain function.^{1–3,6} Definitions of mTBI vary and may include a Glasgow Coma Scale score of 13 to 15, with or without loss of consciousness for less than 30 minutes or posttraumatic amnesia lasting less than 24 hours.^{2,4,7–9} Sports-related concussions (SRCs) are defined as a TBI induced by biomechanical forces.¹ Per the October 2016 Consensus Statement on Concussion in Sport, “SRC may be caused either by a direct blow to the head, face, neck or elsewhere on the body with an impulsive force transmitted to the head. SRCs typically result in the rapid onset of short-lived impairment of neurological function that resolves spontaneously. However, in some cases, signs and symptoms evolve over a number of minutes to hours. SRCs may result in neuropathological changes, but the acute clinical signs and symptoms largely reflect a functional disturbance rather than a structural injury and, as such, no abnormality is seen on standard structural neuroimaging studies. SRC results in a range of clinical signs and symptoms that may or may not involve loss of consciousness. Resolution of the clinical and cognitive features typically follows a sequential course. However, in some cases symptoms may be prolonged.¹ The clinical signs and symptoms cannot be explained by drug, alcohol, or medication use, other injuries (such as cervical injuries, peripheral vestibular dysfunction, etc.) or other comorbidities (eg, psychological factors or coexisting medical conditions).” This imprecise definition and lack of standardized diagnostic criteria contribute to variability in diagnosis clinical practice, which can confound comparative analysis of available evidence.^{2,10} Consequently, the term “concussion” is used throughout this paper and is inclusive of concussion and mTBI. Diagnosis of concussion can also be complicated by the range of self-limited, nonspecific, self-reported symptoms (most commonly headache and dizziness) that may resolve prior to initial presentation or be delayed in onset.^{1,11} The symptoms of concussion are thought to be mostly a result of a neurometabolic cascade of events triggered by the initiating biomechanical forces transmitted to the brain.^{3,5,6} This cascade increases local energy demand in the presence of decreased cerebral blood flow and mitochondrial dysfunction. Although

this pathophysiology is not usually a cause of significant neuronal cell death, it may lead to decreased neural responsiveness, manifesting as persisting symptoms and increasing vulnerability to further dysfunction if overstressed before full recovery.^{3,6}

The incidence of concussion in the United States has been estimated to be 100 to 300 cases annually per 100,000 persons. However, because it has also been estimated that 45%–65% of children and 42% of adults do not access health care after concussion and the injuries go unreported, the actual incidence may be as high as 600 cases per 100,000 persons.^{8,12} Significant populations at risk of concussion include athletes, military personnel, and persons involved in motor vehicle accidents.⁸ An estimated 3.8 million recreation- and sport-related concussions occur annually in the United States, of which 1.1–1.9 million occur in children 18 years of age or younger.³ Although the overall incidence of concussion is higher among male athletes, the incidence in female athletes is nearly twice that of male athletes when comparing rates for equivalent sports.⁸ Among high school and college athletes, male football players and female soccer players have been suggested to have the highest rates of concussion,^{3,8} although these findings may vary by study. The incidence of concussion in deployed U.S. military personnel has been estimated to be as high as 22%, with the highest risk associated with younger age, male gender, White race, lower military rank, Army or Marine Corps service, combat experience, and direct occupation support.^{8,13} The reported incidence of concussion has been increasing, which has been attributed to increasing awareness and education among coaches, medical professionals, and the lay public.^{3,6,13} Despite better awareness and education, some populations of patients continue to receive disparate care.¹⁴ For example, Black adolescents have reportedly higher concussion rates than White adolescents but lower diagnosis, health care utilization, and follow-up rates in addition to worse clinical outcomes after concussion.^{15,16} Among high school athletes, Black athletes have also been reported to be less knowledgeable about concussion signs and symptoms than White athletes,¹⁶ which may prevent them from seeking assessment and care. Improving access to education and health care resources for all populations may lead to more equitable care and outcomes in concussion management.^{16,17}

Outcomes of concussion vary and range from full, rapid recovery to chronic neurologic dysfunction and other sequelae.⁷ In children and adolescents, sports-related concussions have repeatedly demonstrated significant impacts on academic performance.⁵ Among

athletes and former athletes, concussions are associated with an increased risk of depression, impaired balance, mild cognitive deficits, and long-term psychiatric and psychological sequelae, especially in athletes with a history of multiple concussions.^{10,18,19} Posttraumatic stress disorder, physical health problems, and depression have been reported in military personnel several months after concussion, representing a persisting condition.²⁰ Persistent postconcussion symptoms, which are referred to as “persisting symptoms after concussion” in this paper, include a variety of physical symptoms, neurocognitive deficits, and psychiatric conditions, including headache, dizziness, fatigue, auditory deficits, depression, anxiety, irritability, blurred vision, nausea, double vision, confusion, and noise and light sensitivity.^{4,7,21,22} An estimated 15%–20% of patients with concussion develop persisting symptoms after a concussion, but the incidence is likely higher according to objective assessments of cognitive function, which have shown residual cognitive deficits in some patients whose clinical symptoms have resolved.^{4,5,23} Persisting symptoms after a concussion may result in long-term cognitive and physical impairment, adverse mental health conditions, and decreased quality of life, representing a significant health care burden.^{7,22}

However, persisting symptoms after a concussion are nonspecific, and assessing for preexisting, coexisting, and resulting comorbidities is essential. These persisting symptoms may be due to the concussion itself but are often attributable to other factors as well. As an example, preexisting mood disorder and high symptom load immediately after sports-related concussion have been shown to be the most consistent predictors of persisting symptoms in youth athletes. Many patients with persisting symptoms have treatable conditions.²⁴

Other reviews have identified multiple modalities for treatment of concussion and postconcussion symptoms. These include cognitive training, psychoeducation, vestibular rehabilitation, cervical spine therapy, multidisciplinary rehabilitation, graded physical activity, and bed rest. Results from studies evaluating the various treatment modalities are mixed, which limits the conclusions that can be drawn. In general, graded physical activity, psychoeducation, and cognitive training had potential as effective therapies for concussion by some reviewers,^{25,26} whereas others concluded that rest is underused and more evidence is needed to determine the effectiveness of graded return to physical activity.²⁷ Strict rest and intense physical activity were found to be either ineffective or detrimental with respect to symptomatic improvement or neurocognitive assessment in children in randomized controlled trials (RCTs) despite demonstrating a beneficial effect in small retrospective studies.^{26–28} A scoping review of nonpharmacological treatments for persistent postconcussion symptoms found only weak evidence for these therapies, including graded physical exercise, vestibular rehabilitation, and psychological treatments.²⁹ Current

reviews do not provide clear evidence regarding appropriate treatment modalities for concussion.

Current clinical guidelines focus on early diagnosis and management of concussion, which are critical for preventing poor outcomes and minimizing health care burden but provide limited guidance on the use of specific therapeutic interventions, including rest, rehabilitation, and pharmacotherapy, due to insufficient evidence on the effectiveness and appropriateness of these interventions.^{1,13,21,30} Although management of concussion is inclusive of all interventions, the authors sought to determine which interventions are more effective and thus specific interventions were selected. The purpose of this project was to systematically review the available evidence for four categories of therapeutic interventions for concussion and develop an evidence-based consensus statement regarding the use of these interventions in clinical practice.

METHODS

Work group selection process

In 2016, the American Academy of Physical Medicine & Rehabilitation (AAPM&R) held a Concussion Think Tank composed of several physician experts in concussion and TBI. The group recommended AAPM&R convene a multispecialty panel to explore management of concussion. In 2018, AAPM&R contacted several medical associations with established expertise in concussion management to participate in a Concussion Writing Panel, charged with developing an evidence-based consensus statement on the treatment of concussion. The invited specialty societies each selected a member with clinical experience in concussion to represent their specialty. The following specialties were represented on the panel: American Academy of Family Physicians, American Academy of Neurology, American Association of Neurological Surgeons, American Academy of Pediatrics, American College of Sports Medicine, American Medical Society for Sports Medicine, and American Orthopaedic Society for Sports Medicine. In addition, AAPM&R identified a member to serve as the panel chair and three additional PM&R physician members to serve on the 10-member panel.

Scoping review and clinical question development

A scoping review was performed by searching online databases with general search terms and keywords related to the treatment of concussion to identify therapeutic areas of significant interest in the scientific literature. From this review, the panel decided to focus on the efficacy of the nonpharmacologic interventions rest,

TABLE 1 PICO and General Clinical Questions.

Intervention	PICO question	General clinical question
Rest	In patients with concussion, does rest vs. no/other intervention improve outcomes?	Does rest improve outcomes for patients with concussion?
Exercise	In patients with concussion, does exercise therapy vs. no/other intervention improve outcomes?	Does exercise improve outcomes for patients with concussion?
Rehabilitation	In patients with concussion, does rehabilitation vs. no/other intervention improve outcomes?	Does rehabilitation improve outcomes for patients with concussion?
RTA protocol	In patients with concussion, do RTA protocols vs. no/other intervention improve outcomes?	Do RTA protocols improve outcomes for patients with concussion?

Abbreviations: PICO, population, intervention, comparator, outcome; RTA, return to activity.

exercise, rehabilitation, and return to activity (RTA) protocols. These interventions were chosen because they are commonly used, discussed, and studied, and determined to be of high priority by the Concussion Writing Panel. The panel also discussed education as an intervention, but ultimately it was not included due to lack of literature for consideration. Rest, as opposed to no intervention, consisted of directed cognitive, physical, or combined cognitive and physical rest. For each intervention, the panel developed a specific PICO (population, intervention, comparator, outcome) question for systematic review purposes and a general clinical question for broader relevance to clinical practice and to allow for noncomparative studies to be included in the review (Table 1).

Systematic literature search and review

A systematic literature search and review was performed between 2018 and 2019 according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines³¹ to identify relevant original research for each PICO question. Due to a delay related to the COVID-19 pandemic, the same search was repeated from the end of the initial search through June 2022 to identify any additional relevant literature. Electronic searches of PubMed were performed for each PICO question by using appropriate combinations of MEDLINE Medical Subject Headings terms and keywords and by setting available database filters for publication type (various; see inclusion/exclusion criteria), population (humans), language (English), and publication date (January 1, 1990–July 5, 2022), as listed in the [Appendix](#).

Study selection for each PICO question was performed with a two-level screening process. Level 1 screening was performed by three independent reviewers and involved a review of the study titles and, if needed, abstracts to determine initial relevance to the PICO questions. Level 2 screening was performed by one primary reviewer and involved review of the full-text articles to confirm relevance and compare study details with the inclusion/exclusion criteria and definitions described in the [Appendix](#). To summarize, studies were included if they involved patients

5 years or older with head injuries consistent with acute or chronic concussion, were prospective or retrospective original research, and evaluated an association between the intervention of interest (rest, exercise, rehabilitation, RTA protocol) and outcomes after concussion. Studies were excluded if they included children younger than 5 years of age; had mixed populations of brain injury that were not evaluated separately through subanalyses; were nonoriginal research (review articles, systematic reviews/meta-analyses) or editorials/commentaries; or evaluated interventions outside of the scope of this project.

Duplicate articles were eliminated. Excluded studies and their reasons for exclusion were documented for review by the panel to confirm the final rejection or to reconsider the study for inclusion. Final included and excluded studies are tabulated in a PRISMA flow chart for each PICO question ([Appendix](#)). Additional references were included in this review if considered necessary for background or discussion; however, these references were not critically appraised for study quality or included in the consensus voting process.

Data extraction and quality analysis

Studies meeting inclusion criteria underwent data extraction and critical appraisal of methodologic quality. Quantitative and qualitative data relevant to the clinical questions were extracted and compiled for review by the Panel (Table 2). Study quality was assessed with critical appraisal tools developed by the Centre for Evidence-Based Medicine and Cochrane Collaboration.^{32–34} Each study was critically appraised by one primary reviewer and subsequently cross-checked by two panel members. Studies were categorized according to their risk of bias (low-high).

Categorization of studies for clinical questions

To minimize confounding when interpreting the study findings, studies for each overarching clinical question

TABLE 2 Study details, findings, and risk of bias for included studies.

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Rest						
Brooks 2016	Prospective comparative study	154 children/adolescents with mTBI Mean age, ~13 years; 58.4% male in each group Source: ED of tertiary care pediatric hospital	Brief cognitive testing in ED vs. no cognitive testing	Change in PCSI score at 7–10 days, 1 month, 2 months, and 3 months post-injury	No significant change in PCSI at all time points ($F [2.739, 388.939] = 1.90$; $p = .14$).	Moderate
Brown 2014	Prospective cohort with multivariate analysis	335 children/adolescents/adults with sport-related concussion (or similar mechanism, eg, fall) Mean age, 15 years; 62% male Source: Boston Children's Hospital	Self-reported cognitive activity (scale, 0–4; 0 = rest, 4 = full activity), converted to cognitive activity-days (average cognitive activity level $\times$ days between visits), divided into 4 quartiles	Symptom duration: PCSS	Cognitive activity level was independently associated with symptom duration, with lower levels of cognitive activity resulting in statistically shorter symptom duration vs. the highest level of activity with multivariate Cox proportional hazards modeling (HR, .9942; 95% CI, 0.9924–0.9960)	Moderate
Buckley 2016	Retrospective comparative	50 college athletes (adolescent age) with sport-related concussion Mean age: ~19 years in each group; 72% male in rest group, 52% male in no rest group Source: University of Delaware, Sports Medicine Dept	Brief (~40 h) of cognitive/physical rest vs. no rest	Number of days until symptom free Number of days until baseline values were met or exceeded on the CNT, BESS, SAC Time until clinical recovery	A prescribed 1 day of cognitive and physical rest was associated with a longer duration of symptoms (5.2 ± 2.9 days vs. 3.9 ± 1.9 days; $p = .047$). No change in time to baseline BESS (2.7 ± 1.9 days vs. 2.1 ± 1.3 days; $p = .183$), SAC (2.6 ± 1.9 days vs. 2.1 ± 1.5 days; $p = .368$), CNT (5.7 ± 5.0 days vs. 7.1 ± 6.3 days; $p = .395$) with rest vs. no rest. No change in time to clinical recovery for rest vs. no rest (6.8 ± 4.6 days vs. 7.2 ± 5.8 days respectively; $p = .809$).	Moderate
de Kruijk 2002	RCT	107 adolescents/adults with mTBI Mean age, ~34–39; ~52%–60% male, depending on study group Source: Maastricht University Hospital	6 days of bed rest vs. no bed rest	Posttraumatic complaints: VAS	No significant differences in outcomes between the rest and no rest groups for any of the follow-up periods ($p > .05$).	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Gibson 2013	Retrospective observations (similar to a case control study)	184 children/adolescents/adults with sport-related concussion Mean age, 15 years; 72% male Source: Sports Concussion Clinic; Children's Hospital	Recommendation of cognitive rest	Association between cognitive rest recommendation and symptom duration Patients assigned based on the outcome duration of symptoms: Patients with symptoms >30 days Patients with symptoms ≤30 days	The recommendation for cognitive rest was not significantly associated with duration of symptoms (adjusted OR, 0.5; 95% CI = 0.18–1.37)	High
Grool 2016	Prospective cohort/comparative	2413 children/adolescents with concussion in preceding 48 h Mean age, 11.77 years; 60.7% male Source: Pediatric Emergency Research Canada network tertiary pediatric EDs	Early physical activity within 1 week of injury vs. no physical activity	Development of persisting symptoms postconcussion (28 days postinjury): persistent postconcussion syndrome, PCSI	Risk of persistent postconcussion syndrome was lower in the physical activity group than the no physical activity group (relative risk, 0.75; 95% CI, 0.70–0.80; absolute risk reduction, 18.9%; 95% CI, 14.7%–23.0%). All classifications of exercise (light, moderate, or full) were associated with lower risk of persistent symptoms	Moderate
Howell 2016	Prospective cohort/comparative	364 children/adolescents/adults with sports-related concussion or by similar mechanism (falling, injured during recreational activities) Mean age, 15 years; 61% male Source: sport concussion clinic	Self-report of physical activity level (1 = rest; 7 = full game play) and cognitive activity	Symptom duration: PCSS	Cox regression analysis demonstrated that physical activity was associated with shorter symptom duration for adolescents.	Moderate
Maertender 2015	RCT	28 college athletes with mTBI/sports-related concussion Mean age not reported but like adolescent age range; 28.5% male Source: University of Nebraska	Prescribed daily physical exertion (stationary bike at mild/moderate level Borg CR10 RPE scale) vs. standard concussion recovery recommendations (limited exertion to normal activities required for school)	Time to recovery, when test scores returned to baseline: ImPACT, neurocognitive test battery, post-ride symptom change rating	No significant difference in time to recovery between the exertion and standard groups (15 days vs. 13 days; $p = .705$). More vigorous average daily activity adversely accounted for variance in recovery time (mean 7.02; $P = .039$)	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Moser 2012	Retrospective case series with stratification by time between injury and assessment	49 adolescents/adults with concussion Mean age, ~14–15 years; ~18%–46% male, depending on group Source: Sports Concussion Center of New Jersey	Prescribed at least 1 week of cognitive and physical rest	Change in symptoms and performance: ImPACT	Physical and cognitive rest was associated with significantly improved ImPACT performance ($F [10, 86] = 1.02; p < .001$). Time between concussion and rest onset was not associated with ImPACT performance	High
Moser 2015	Retrospective case series	13 adolescents with concussion and persisting symptoms Mean age, 15.1 years; 57% male Source: specialty clinic	Prescribed rest for 1 week	Change in symptoms and cognitive function: Total Symptom Score and ImPACT	Analysis of variance demonstrated improvement in symptoms and cognitive functioning after receiving education, reassurance, and 1-week of prescribed rest	High
Root 2020	Prospective cohort study	119 children/adolescents with concussion Mean age 10.4–11.6 years; 45%–52% male Source: pediatric ED or a satellite community ED	Less than 2 days of cognitive or physical rest vs. 2 days or more of cognitive or physical rest	PCSI obtained 28–32 days after injury	Moderate cognitive rest had a higher adjusted OR of persistent postconcussive symptoms than minimal cognitive rest (1.15; 95% CI, 0.44–2.99). Moderate physical rest had a higher adjusted OR of persistent postconcussive symptoms than minimal physical rest (3.17; 95% CI, 0.35–28.78).	High
Sufrinko 2017	Prospective cohort/comparative (secondary analysis of Thomas 2015)	93 adolescents with concussion Mean age, 13.87 years; 64.5% male Source: ED	Rest vs. usual care Patients categorized into “signs” or “symptoms” groups according to the presence of clinical signs and subjective symptoms by PCSS before intervention	Change in neurocognitive performance: BESS, ImPACT	When prescribed rest, those in the “symptoms” group had a significantly higher likelihood of persistent symptoms at days 1–6 and 8 (range of ORs 3.56 to 5.82; $p < .03$). Prescribed rest improved verbal memory performance in the “signs” group ($t = -2.28, P = .029$) but had no effect on the “symptoms group”	Moderate
Taubman 2016	Prospective cohort/comparative	95 adolescents with concussion Mean age, 14.3 years; 60% male Source: private pediatric practice	Immediate cognitive and physical rest (same day as injury) vs. delayed cognitive and physical rest (1–7 days after injury)	Recovery from the concussion: ability to return to school full-time or do unlimited cognitive work	Immediate cognitive and physical rest was associated with an increased rate of recovery within 30 days compared with delayed rest (67% vs. 35%, $p = .016$). Within the subset of participants who recovered within 30 days, those who	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Thomas 2015	RCT	99 adolescents with concussion Median age, 13.7; 66% male Source: Children's Hospital of Wisconsin ED and Trauma Center	5 days of strict rest (intervention) vs. usual care of 24–48 h of rest (control)	Change in postconcussion symptoms and neurocognitive and balance assessment scores: PCSS, ImPACT, BESS, and ancillary neurocognitive assessments	underwent immediate cognitive and physical rest recovered 4.6 days faster (10.29 ± 5.83 vs. 14.42 ± 6.15 days; $P = .005$) There was no significant difference in the rate of symptom resolution with strict rest compared with control (63% vs. 67%; $p = .82$). For those prescribed strict rest, 50% required 3 additional days before symptom resolution. Strict rest was associated with a higher total PCSS (187.9 vs. 131.9; $p = .03$) and higher total number of postconcussive symptoms (70.4 vs. 50.3; $p < .03$) than control.	Moderate
Varner 2017	RCT	118 adolescents/adults with "head injury" Mean age 35.2 years; 36.4% male Source: ED of tertiary care center	Discharge instructions for cognitive rest and graduated RTA (intervention) vs. usual care, comprised of discharge instructions about concussion symptoms and warning signs of a missed intracranial injury (control)	Change in symptoms, PCSS	Change in PCSS was not statistically different between the treatment group and control group (−10.5 vs. −12.8; difference, −2.3; 95% CI for difference, −11.7 to 7.0). No differences in rate of missing work/school, numbers of work/school days missed, or number of visits to health care provider	Moderate
Willer 2019	Prospective cohort study	56 adolescents with sports-related concussion compared with 113 participants in historical control cohorts Mean age 15.4 year; 75% male Source: University-affiliated concussion clinics	Relative rest was recommended. Compared with historical cohorts undergoing active control (stretching) and aerobic exercise	Days to recovery from injury (PCSS less than 7 for 3 consecutive days)	Time to recovery was longer in the rest group compared to the control group (median 16 days; IQR, 9.25–23.25 vs. 13 days; IQR, 10–18.7] $p = .020$).	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Exercise						
Buckley 2022	Prospective observational study	78 collegiate athletes with sports-related concussion Mean age 19.6 years; 51.3% male Source: National Collegiate Athletic Association Division 1 athletes and cheerleaders	Self-reported physical and mental activity scales in the first 2 days after concussion	Number of days until full return to play and symptom resolution	Mild physical activity in the first 2 days after concussion was a significant predictor of shorter time to being symptom free ($p = .002$) and faster RTA ($p = .006$).	High
Chan 2018	RCT	19 adolescents with sport-related concussion with persisting symptoms Mean age, 15.5; 26% male Source: Tertiary outpatient concussion clinic	Active rehabilitation vs. treatment as usual for 6 weeks	Change in self-reported symptoms: PCSS	Active rehabilitation was associated with a greater reduction in mean PCSS than treatment as usual (-24.7 ± 19.1 vs. -15.8 ± 12.5 ; Cohen's d treatment effect size, .55)	Moderate
Chrisman 2017	Retrospective case series, stratified by duration of persistent symptoms at presentation	83 children/adolescents with concussion and persisting symptoms Mean age, 14.9 years; 46% male Source: Seattle Children's Hospital	SSTEP program at least 1 week	Change in symptoms: SCAT-2	Symptoms improved exponentially over time and decreased by 45% from baseline to 30 days, with similar improvements irrespective of symptom duration at baseline	High
Cordingley 2016	Retrospective case series	106 children and adolescents with sport-related concussion; 58 with persisting symptoms Mean age 15.1 years; 43.4% male Source: Pan Am Concussion Program	BCTT and clinical exam to diagnose symptoms Submaximal exercise program for 5 days/week	Change in symptoms: PCSS Clinical outcomes after submaximal program: clinically improved, not clinically improved, incomplete response	Submaximal exercise was safe and well tolerated in patients with postconcussive symptoms. With submaximal exercise, 81.0% of those with postconcussive symptoms showed clinical improvement	High
Coslick 2020	Retrospective cohort study with regression analysis	178 adolescents with concussion Mean age 13.7 years; 50% male Source: academically affiliated rehabilitation concussion clinic	Self-reported physical activity was dichotomized into 2 groups consisting of low activity level and high activity level	Days to recovery from concussion event based on clinician assessment Other outcomes included ACE and response speed, memory/learning, attention, and executive functioning assessment	Moderate to heavy physical activity was associated with a shorter time to recovery than light or no activity (35.1 days vs. 59.1 days; 95% CI for difference -36.3 to -11.9 ; $p < .001$). Results persistent after controlling for confounding factors (difference, -21 days; 95% CI, -27.07 to -15.48 ; $p < .001$)	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Dobney 2017	Retrospective case series	227 children/adolescents with concussion Mean age, 14.1 years; 51.3% male Source: mTBI program	Active rehabilitation (aerobic exercise with a stationary bike, coordination drills, visualization, and education)	Change in symptom scores: PCSS, SCAT-2, and SCAT-3	Physical activity was associated with a decrease in total severity score from median 18 preintervention to 9.5 postintervention ($p < .05$). Improvement was seen in physical, cognitive, emotional and sleep clusters	High
Gagnon 2016	Prospective case series	10 adolescents with sports-related concussion, slow to recover Mean age, 16.3 y; 70% male Source: Concussion Clinic of the Montreal Children's Hospital	Rehabilitation (submaximal aerobic training, light coordination exercise, visualization/imagery, and individualized home exercise) for at least 6 weeks	Change in symptoms: PCSS	Postconcussion symptoms and functioning significantly improved after 6 weeks of the active rehabilitation program (PCSS 17.4 preintervention to 2.7 postintervention; $p = .004$)	High
Grabowski 2017	Retrospective case series	25 adolescents with concussion and persisting symptoms Mean age, 15 years; 44% male Source: University of Wisconsin Hospitals and Clinics	Impairment-based therapy involving subsymptom threshold exercise and comprehensive physical therapy 3 core components (1) a subsymptom cardiovascular exercise protocol, (2) vestibular/OMT exercise, and (3) cervicothoracic manual therapy and therapeutic exercise	Change in symptoms: PCSS	Median PCSS scores decreased significantly from baseline to postintervention follow-up (18.2 vs. 9.1; $p < .01$)	High
Grool 2016	Prospective cohort/comparative	2413 children/adolescents with concussion in preceding 48 h Mean age, 11.77 years; 60.7% male Source: Pediatric Emergency Research Canada network tertiary pediatric EDs	Early physical activity within 1 week of injury vs. no physical activity	Development of persisting symptoms postconcussion (28 days postinjury): PPCS, PCSI	Rate of persisting symptoms was significantly lower with early physical activity vs. no physical activity (relative risk, 0.75; 95% CI, 0.70–0.80; absolute risk reduction, 18.9%; 95% CI, 14.7%–23.0%)	Moderate
Howell 2016	Prospective cohort/comparative	364 children/adolescents/adults with sports-related concussion or by similar mechanism (falling, injured during recreational activities) Mean age, 15 years; 61% male Source: sport concussion clinic	Self-report of physical activity level (1 = rest; 7 = full game play) and cognitive activity	Symptom duration: PCSS	Symptom duration was similar between activity level groups in the overall study population, but adolescents with lower levels of physical activity appeared to have longer symptom duration	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Howell 2020	Prospective cohort study	72 adolescents with concussion Mean age 20.2 years; 54% male Source: National Collegiate Athletic Association Division I university	Subsymptom threshold exercise at the discretion of independent medical staff vs. no exercise	PCSS at initial and follow-up evaluations, domain-specific symptoms severity ratings, single/dual-task gait assessment, and modified BESS	Post-injury exercise was not associated with a difference in median PCSS (18.5; IQR, 7.5–26 vs. 17; IQR 14–40; $p = .21$), median days to resolution (13; IQR, 7–18 vs. 13; IQR, 7–23 days; $p = .83$), or median days to clearance to play (15; IQR, 11.5–36 vs. 19; IQR, 14–26; $p = .46$)	Moderate
Howell 2021	RCT	37 adolescents with concussion Mean age 16.8–17.2; 50%–59% male Source: Children's hospital affiliated sports medicine clinic or ED	Heart rate-targeted exercise program 4 weeks vs. no specific exercise recommendation	PCSI and diary of exercise assessed weekly	Exercise compared with control was not associated with change in initial median PCSI (15; IQR, 10–42; vs. 20; IQR, 11–35; $p = .26$) or PCSI at 1 month (4; IQR, 4–28 vs. 5.5; IQR, 0.5–21.5; $p = .11$). Exercise volume was similar between groups	High
Howell 2022	Pilot RCT	16 adolescents with concussion at risk for persistent postconcussion syndrome Mean age 13.6–14.2 years; 43%–44% male Children's Hospital of Colorado ED and sports medicine clinic	Individualized aerobic exercise recommendation for 1 month with target heart rate based on symptoms vs. standard of care with no specific recommendation	Presence of persistent postconcussion syndrome at 1 month, PCSI, compliance with exercise intervention	Early exercise program was associated with decreased but nonsignificant risk of developing persistent postconcussion syndrome (risk ratio, 0.52; 95% CI, 0.34–1.36; number needed to treat, 2.4)	High
Kurowski 2017	RCT	30 adolescents with mTBI and prolonged symptoms Mean age, ~15 years; ~33%–53% male, depending on group Source: community, brain/head injury clinics, and EDs	Cycling intervention (Borg intensity range of 11 to 16) vs. full-body stretching intervention for 6 weeks	Improvement in postinjury symptom improvement: PCSI	Subsymptom aerobic training significantly improved symptoms over time vs. full-body stretching in repeated-measures mixed-models analysis ($F = 4.11$, $p = .044$)	Moderate
Lawrence 2018		253 adolescents with acute concussion	Timing of self-initiated or physician-prescribed	Time (days) from injury to full return to sport and	Exercise initiation on 3, 5, 7, and 14 days post-injury was	Moderate (Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
	Retrospective cohort/comparative	Median age, 17 years; 58.5% male Source: academic sports medicine clinic	aerobic exercise after injury (within 1 day vs. at 3 days)	school/work, defined as time to full return as assessed by sports physician or to physician clearance for full return	associated with a 36.5% (HR, 0.63; 95% CI, 0.53 ± 0.76), 59.5% (HR, 0.41; 95% CI, 0.28 ± 0.58), 73.2% (HR, 0.27; 95% CI, 0.16 ± 0.45), and 88.9% (HR, 0.11; 95% CI, 0.06 ± 0.22) reduced probability of faster full RTA, respectively	
Larson-Dupuis 2021	Prospective cohort study	35 adults, 18 with concussion Mean age 58.9 years; 29% male Source: concussion patients recruited from a level-1 trauma center. Healthy controls were recruited from newspaper and online advertisements	Aerobic exercise program vs. stretching program	Neuropsychological assessment including Conners Continuous Performance Test II, Rey Auditory Verbal Learning Task, Brief Visual Memory Test – Revised, Block Design Test of the Weschler Adult Intelligence Test-III, Tower of London, D-KEFS, processing speed testing, Beck Anxiety Inventory, Beck Depression Inventory II, and cardiopulmonary exercise testing	Participants in the aerobic exercise program did not demonstrate improvement in measures of cognition compared with those in the stretching program	High
Leddy 2010	Prospective case series	13 adolescents/adults with persisting symptoms after concussion Mean age, 27.9 years; 58% male Source: University at Buffalo Concussion Clinic	Subsymptom threshold exercise	Safety and effectiveness Improvement in persisting symptoms: GSC	Subsymptom exercise led to an increase in exercise time from 9.78 to 18.67 min ($p = .001$), peak heart rate (147–179 beats per minute; $p = .001$), and systolic blood pressure (142–156 mmHg; $p = .001$) without symptom exacerbation	High
Leddy 2013	Prospective cohort/comparative	12 children/adolescents, 8 with concussion and persisting symptoms (4 healthy controls included, which were not evaluated for this review) Age range of concussion patients, 17–52 years; 20%–50% male. Source: university concussion clinic and the community	Progressive aerobic exercise program vs. placebo stretching program	Changes in brain activity (fMRI with math processing test), exercise heart rate, symptoms (PCSS)	Exercise, when compared with a stretching control, was associated with a higher mean tolerated heart rate (185 vs. 152 beats per minute; $p < .001$) and fewer symptoms (2 vs. 15; $p < .0004$)	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Leddy 2019	Prospective cohort study	67 adolescents with sports-related concussion Mean age 15.1–15.5 years; 100% male Source: 3 university concussion management clinics	BCTT-guided aerobic exercise program vs. matched historical cohort instructed not to participate in physical activity	PCSS, time to recovery from concussion	Time to recovery was faster for those in the exercise group than those in the historical control group (8.29 ± 3.85 days vs. 23.93 ± 41.73 days, $p = .048$). Lower rate of prolonged symptoms in the exercise group than control group (8% vs. 33%, $p = .028$)	High
Leddy, Haider, and Ellis 2019	Multicenter RCT	103 adolescents with sports-related concussion Mean age 15.3–15.4 years; 53%–54% male Source: 3 university-based outpatient concussion clinics	BCTT-guided aerobic exercise program vs. stretching program	Time to recovery, portion of participants with prolonged recovery, daily symptom scores	Aerobic exercise was associated with a faster time to recovery than the stretching program (median 13 days; IQR, 10–18.5 days vs. 17 days; IQR, 13–23 days; $p = .009$). Daily symptom scores were similar in both groups.	Low
Leddy 2021	Multicenter RCT	118 adolescents with sports-related concussion Mean age 15.5–15.9; 62%–63% male Source: University-affiliated sports medicine clinic/children's hospital-affiliated clinics	BCTT-guided aerobic exercise program vs. stretching and breathing exercise	PCSI, symptom reporting, heart rate data, time to recovery	Aerobic exercise had effect on clinical recovery at 4 weeks (HR, 0.52; 95% CI 0.28–0.97; $p = .039$). Per protocol analysis demonstrated reduced median days to recovery in the exercise group compared with control (12 days; IQR, 10–22 days vs. 21 days; IQR, 13–35 days; $p = .011$). Reduced rate of postconcussive symptoms in the exercise group compared with control (9% vs. 31%; $p = .014$)	Moderate
Maerfender 2015	RCT	28 college athletes with mTBI/sports-related concussion Mean age not reported but like adolescent age range; 28.5% male Source: University of Nebraska	Prescribed daily physical exertion (stationary bike at mild/moderate level Borg CR10 RPE scale) vs. standard concussion recovery recommendations (limited exertion to normal activities required for school)	Time to recovery	No significant difference in median days until recovery between the exertion and standard groups (15 vs. 13 days; $p = .705$)	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Majerske 2008	Retrospective cohort/comparative	95 children/adolescent student-athletes with sport-related concussion Mean age, 15.88 years; 84% male Source: University hospital sports medicine concussion program	Patients grouped according to self-reported activity on Activity Intensity Scale (0 = no school or exercise activity, 4 = school activity and participation in a sports game)	Change in symptoms and function: PCSS, ImPACT	Level of exertion was significantly related to all outcome variables ($p < .02$ for all comparisons); moderate levels of activity were associated with better symptom and functional improvements, whereas higher levels of activity (activity intensity scale of 3 or 4) associated with the worse visual memory scores ($p < .05$ for all comparisons)	Moderate
Moore 2016	Prospective case series, with age stratification	14 adolescents/adults with persisting symptoms and disability after concussion Median age, 43 years; 43% male Source: New York University Langone Medical Center Concussion Program and Vestibular Rehabilitation Clinic	Supervised home exercise program consisting of vestibular rehabilitation combined with aerobic training for 6 months	Change in symptoms and disability: RPQ-3, RPQ-13, DHI, ABC, and FGA, and ability to RTW, RTA	Statistically significant improvements in symptoms and disability were observed between baseline and follow-up measurements at 3 months (RPQ-13 [$p < .002$]; DHI [$p < .001$]; and FGA [$p < 0.01$]) and at 6 months (RPQ-3 [$p < .001$]; RPQ-13 [$p < .004$]; DHI [$p < .001$]; ABC [$p < .007$]; FGA [$p < .001$])	High
Popovich 2021	Retrospective cohort study	108 adolescents with sports-related concussion Mean age 15.3–15.4; 69.4%–76.9% male Source: Michigan NeuroSport Clinic	BCTT-guided subsymptom exercise vs. no exercise/delayed exercise	Safety events, days until return to sport, symptomatic days	Exercise was associated with early return to sport (HR, 2.35; $p = .03$) and lower mean days from injury to return to sport compared with control (26.5 vs. 35.1 days; $p = .02$). No safety events were identified	High
Remigio-Baker 2020	Prospective cohort study	62 adult military service members with concussion Mean age 26.3 years; 88.7% male Source: 3 military installations	Assessments over 6 months to assess activity level impact on symptoms	60-item cognitive, lifestyle, physical, vestibular/balance, military-specific, and rest Neurobehavioral Symptoms Inventory cognitive, vestibular, somatosensory, and affective items	High activity level was associated with increased symptoms for participants who had higher acute concussion symptoms. No relationship was found for those with lower baseline symptoms	High
Seehusen 2021	Prospective cohort study	32 adolescents with concussion Mean age 14.9–15.4 years; 53%–60% male Source: Children's Hospital of Colorado	Exercise tracker monitored activity for participants who returned to play within 28 days vs. those that returned to play after 28 days	Steps per day, workouts per week, workout duration, total exercise time per week, heart rate during exercise, PCSI	Participants who returned to activity after 28 days compared with those who returned sooner had higher mean PCSI (70.2; SD, 22.3 vs. 42.9; SD, 24.9; $p = .003$).	High

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Snyder 2021	Pilot RCT	25 adolescents and adults with concussion Mean age 20.4–22.2 years; 46.2%–60% male Source: Neurotrauma ED or Sports Concussion Center	Daily aerobic exercise guided by estimated maximum heart rate vs. stretching and calisthenic movement program	SCAT-3, Sleep Problems Index II, Beck Depression Inventory, State Trait Anxiety Inventory, modified BESS, 2-hour neurocognitive batter	Similar findings across physical, cognitive, emotional, and fatigue domains. Those that had RTP after 28 days had significantly lower steps per day ($p = .018$), exercise sessions per week ($p = .013$), and exercise volume ($p = .03$) No difference in symptom severity or outcomes between aerobic and nonaerobic exercise group (visit by group interaction $p = .78$)	High
Stumph 2020	Retrospective cohort study	187 adolescents with sports-related concussion Age range 10–17; 55% male Source: 7 pediatric hospital-affiliated concussion clinics	Recommendation for subsymptom exercise without contact vs. recommendation to avoid physical activity for at least 4 days	Symptom scores, days from injury to first visit, days from exercise prescription to symptom resolution, days from injury to symptom resolution, days from initial visit to symptom resolution	Median PCSS 40, no significant difference between groups ($p = .68$). Increase in days to recovery in early exercise group compared with delayed activity (18.5 vs. 14 days, $p = .002$)	Moderate
Vamer 2021	RCT	367 adults with concussion Median age 32; 42.4% male Source: 3 Canadian EDs	30 min of light exercise daily vs. cognitive rest	Rivermead Post Concussion Symptom Questionnaire	Exercise did not affect the proportion of participants with persistent postconcussion symptoms compared with control (14.6% vs. 13.4%, 95% CI for difference –6.2 to 8.5) or self-reported symptom score compared with control (13 vs. 14, 95% CI for difference –1 to 4). No change in number of health care visits, unplanned ED visits, or missed school/work days	High
Worts 2022	RCT	30 adolescents with sports-related concussion Mean age 15.8–16.3 years; 64%–68% male Source: US sports medicine clinics	Exercise guided by age predicted maximum heart rate vs. seated rest	Heart rate variability, blood pressure, rating of perceived exertion, King-Devick test, VOMS, PCSS	No treatment effect ($p = .436$), time effect ($p = .064$) or treatment by time interaction ($p = .776$) were observed. No impact on vestibular/ocular testing was observed	High

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Yang 2021	Prospective cohort study	83 adolescents with concussion Mean age 14.2 years; 65% male Source: Referral from school athletic trainer or children's hospital ED	Device monitoring of physical and cognitive activity	PCSS, daily steps, minutes, minutes per day on cognitive activity, minutes of school time	Cox proportional hazard models revealed effect of activity on increased likelihood of earlier symptom resolution (HR, 1.17; 95% CI, 1.02–1.34). Increase in school attendance was associated with earlier recovery (HR, 1.17; 95% CI, 1.05–1.30)	Moderate
Rehabilitation						
Ahluwalia 2021	Retrospective cohort study	23 adolescents with concussion Mean age 16.1; 48.5%–50% male Source: Single institution concussion center	Vestibular rehabilitation services earlier than 30 days after injury vs. after 30 days	PCSS, RTP date, and return-to-learn date,	Early vestibular rehabilitation compared with late treatment was associated with shortened time to RTP (31 days vs. 110 days; $p = .03$) but no impact on return-to-learn (17.5 days vs. 12 days; $p = .09$). Symptom resolution was shorter in the earlier group than the late group (54 days vs. 121.5 days; $p = .02$)	High
Alsalaheen 2010	Retrospective case series	84 children/adolescents/adults with concussion Mean age, 25 years; 33% male Source: Referred between 2006 and 2008 to a tertiary balance center for vestibular rehabilitation after being diagnosed of a concussion	Vestibular rehabilitation: daily exercises, customized to each patient's impairment and functional limitations that related to dizziness, ocular motor function, and gait and balance function.	Change in self-reported dizziness/balance measures and gait and balance performance measures: dizziness severity, ABC, DHI, DGI, FGA, FTSTS, SOT	Interaction between treatment and age was significant for dizziness severity ($p = .005$). No significant difference as seen for adults ($p = .805$).	High
Alsalaheen 2020	Retrospective cohort compared with prospective control	154 adolescents (77 with concussion) Mean age 13.4–15.4 years; 42.9% male Source: not clearly stated	Vestibular physical therapy	VOMS	VOMS scores were improved for those with concussion treated with vestibular physical therapy ($p < .001$)	High
Chan 2018	RCT	19 adolescents with sport-related concussion with persisting symptoms Mean age, 15.5; 26% male Source: Tertiary outpatient concussion clinic	Active rehabilitation vs. treatment as usual for 6 weeks	Change in self-reported symptoms: PCSS	Active rehabilitation was associated with a greater reduction in mean PCSS than treatment as usual (-24.7 ± 19.1 vs. -15.8 ± 12.5 ; Cohen's d treatment effect size, .55)	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Ciuffreda 2008	Retrospective case series	33 adolescents/adults with mTBI and vision symptoms Mean age, 42.3 years Source: Raymond J. Greenwald Rehabilitation Center	Conventional vision therapy paradigms Range of 10–30 visits over 2–8 months	Treatment success defined as marked/total improvement in at least 1 primary oculomotor symptom and at least 1 primary sign	A total of 90% of patients achieved complete or marked improvement in one or more oculomotor signs and symptoms	High
Cooper 2017	RCT	126 service members with persisting symptoms Mean age, ~30–33 years; ~86–100% male, depending on study group Source: consecutive outpatient referrals of active duty (or activated reserve/guard) military SMs	Cognitive rehabilitation for 6 weeks vs. control Arm 1 (control): Psychoeducation and medical care for noncognitive symptoms Arm 2: Computerized cognitive rehabilitation Arm 3: Traditional cognitive rehabilitation Arm 4: Integrated cognitive rehabilitation	Change in PASAT, SCL-90-R, KBCI scores	All 4 treatment groups improved over time on the 3 primary outcome measures, but KBCI outcomes were the only statistically significant improvement observed in traditional and integrated groups vs. control ($p = .01$ for both comparisons)	Moderate
Dharm-Datta 2015	Prospective case series	91 military servicepersons, 21 with mTBI Median age at time of injury, 27 years Source: Neurorehabilitation unit at the Defence Medical Rehabilitation Centre Headley Court, UK military rehabilitation unit	3-week admission for multidisciplinary team assessment and multimodal therapy, up to 16 weeks	Change in measures: MPAL-4, converted to T scores Employment status, VIS Living status (independently, with support, or institutional care)	Outcome measures improved in the overall study group, but improvement in the concussion/mTBI population did not reach the minimal clinically important difference of a 5-point reduction in MPAL-4 score (observed difference 1 pt; $p < .001$)	High
Dobney 2017	Retrospective case series	227 children/adolescents with concussion Mean age, 14.1 years; 51.3% male Source: mTBI program	Active rehabilitation (aerobic exercise with a stationary bike, coordination drills, visualization, and education)	Change in symptom scores: PCSS, SCAT-2, and SCAT-3	Total median symptom severity score was reduced after active rehabilitation compared with before (9.5 vs. 18; $p < .05$)	High
Elmark 2007	RCT	395 adolescents/adults with concussion Mean age, ~32–34 years; ~59%–67% male, depending on study group Source: Sodra Alvsborgs Hospital	Individualized rehabilitation (oral information, counseling, encouragement, and referral for individualized additional therapy, eg, occupational therapist, social worker, physiotherapist, and other medical specialists) vs. no rehabilitation (regular care)	Change in symptoms and life satisfaction: PCSQ, LISat-11	No change in postconcussion symptoms (mean score 5.2 vs. 4.4; not significant) with rehabilitation. Significant improvement in the physical health item of the LISat-11 was observed in the intervention group ($p < .05$) was found	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Gagnon 2016	Prospective case series	10 adolescents with sports-related concussion, slow to recover Mean age, 16.3 years; 70% male Source: Concussion Clinic of the Montreal Children's Hospital	Rehabilitation (submaximal aerobic training, light coordination exercise, visualization/imagery, and individualized home exercise) for at least 6 weeks	Change in symptoms: PCSS	Postconcussion symptoms and functioning significantly improved after 6 weeks of the active rehabilitation program (PCSS 17.4 preintervention to 2.7 postintervention; $p = .004$)	High
Gauvin-Lepage 2020	Prospective cohort study	49 adolescents with concussion Mean age 13.2–14.0 years; 48.5%–38.5% male Source: Montreal Children's Hospital concussion clinic and ED	Multimodal rehabilitation with aerobic activity, coordination and sport-specific activity, mental imagery, education, and home program vs. standard of care consisting of symptom-limited activity, education, academic adaptations, gradual return to school, physical activity restriction	PCSI, Beck Youth Inventory for children and adolescents—Second Edition, Child Behavior Check List, Pediatric Quality of Life Multidimensional Fatigue Scale, Pediatric Quality of Life Generic Module, Balance SCAT-3, Bilateral coordination and balance subtests of the Bruininks-Oseretsky Test of Motor Proficiency-2nd Edition, State Trait Anxiety Inventory, IMPACT cognitive functioning score, Physical Activity Questionnaire for Older Children and the Physical Activity Questionnaire for Adolescents, and Pediatric Quality of Life Health Care Satisfaction Generic Module	No significant interaction between groups for child reported PCSI scores ($p = .33$) or parent reported PCSI scores ($p = .59$).	High
Grabowski 2017	Retrospective case series	25 adolescents with concussion and persisting symptoms Mean age, 15 years; 44% male Source: University of Wisconsin Hospitals and Clinics	Impairment-based therapy involving subsymptom threshold exercise and comprehensive physical therapy 3 core components (1) a subsymptom cardiovascular exercise	Change in symptoms: PCSS	Median PCSS scores decreased significantly from baseline to postintervention follow-up (18.2 vs. 9.1; $p < .01$)	High

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Hammerle 2019	Retrospective cohort study	48 adult military service members with concussion Mean age 33.5; 87.5% male Source: Brooke Army Medical Center Brain Injury Rehabilitation Service physical therapy clinic	protocol, (2) vestibular/OMT exercise, and (3) cervicothoracic manual therapy and therapeutic exercise Vestibular rehabilitation therapy compared with cervical spine proprioceptive training	Self-reported symptoms	Cervical spine proprioceptive training was associated with an increase adjusted OR (30.12; 95% CI, 4.44–204.26; $p = .001$) compared with vestibular rehabilitation	High
Han 2018	RCT	56 adults with chronic mTBI (mean time since injury ~9 years) Mean age, ~40–42 years; ~62%–63% male, depending on group Source: Dallas–Ft. Worth community	SMART versus BHW control, for 8 weeks Vestibular rehabilitation exercises with behavioral management strategies vs. behavior management and stretching/physical activity	Neuropsychological assessments: D-KEFS, FSIQ-2, Beck Depression Inventory-II, and PTSD Check List Stressor-specific MRI scans VOMS, DHI, modified BESS, PCSS	SMART significantly improved neuropsychological function vs. BHW, and benefits observed with SMART persisted at 3 months and reached levels observed in healthy populations based on normative scores ($p < .05$) Vestibular rehabilitation was associated with significant differences in vertical vestibulo-ocular reflex (mean difference –1.629; 95% CI, –3.20 to –0.06) and horizontal vestibulo-ocular reflex (–2.24; 95% CI, –4.01 to –0.48). No significant difference in VOMS, DHI, BESS, or PCSS observed.	Moderate
Kontos 2021	RCT	50 adolescents with concussion Mean age 15.4 years; 38% male Source: concussion specialty clinic				Low
Kleffelgaard 2016	Case Series	4 adults with mTBI and dizziness and balance problems Mean age, 36 years; 50% male Source: outpatient clinic	Group-based vestibular rehabilitation for up to 8 weeks	Standardized self-reported and performance-based instruments: DHI, VSS-SF, VSS-V, VSS-A, RPQ with RPQ-3 RPQ-13, QOLIBRI, HADS, BESS, HiMAT	3 of 4 patients noted improvements in self-reported disability and other outcomes	High
Langevin 2022	RCT	60 adults with concussion Mean age 39 years; 30%–33% male	Cervicovestibular rehabilitation program and aerobic exercise vs. exercise alone	PCSS, Numerical Pain Rating Scale, Neck Disability Index, HDI, DHI, time to return to	Adding cervicovestibular rehabilitation did not result in a group-by-time effect ($p = .62$) or effect on number of symptoms ($p = .30$). A	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Mahncke 2021	RCT	Source: Quebec City concussion or medical clinics 83 adult military service members with chronic concussion Mean age 33.8; 81% male Source: 5 military and VA trial sites	Cognitive training self-administered via computer with telephone-based coaching, vs. coaching and video game control	function, and cervical/ vestibular measures Rey Auditory Verbal Learning Test sum of trials 1–5 and delayed recall, the Ruff Light Trails Test sum of trials 2–10 and delayed recall, Digit Span sum of forwards, backwards, sequencing, Symbol Span. Anti-saccades, flanker, and set-shifting from the EXAMINER battery SF-12 mental component score, Beck Depression Inventory, Post-Traumatic Stress Disorder Checklist–Civilian version, Frontal Symptoms Behavioral Scale, the Cognitive Failures Questionnaire, Neurobehavioral Symptoms Inventory, and MPAI	significant group-by-time interaction was observed for flexion rotation testing, head impulse testing, and cervical spine range of motion ($p < .01$) Cognitive training was associated with improvement in composite neuropsychological testing score compared with active control after training (+6.9 points; 95% CI, 1.0 to 12.7; $p = .025$) and at follow up appointment (+7.4 points; 95% CI, 0.6 to 14.3; $p = .0039$)	High
McCarty 2021	RCT	200 adolescents with sports-related concussion Mean age 14.7 years; 38% male Source: primary care, sports medicine, pediatric neurology, and rehabilitation clinics	Cognitive behavioral therapy, care management, and enhanced medication consultation vs. usually care	HBI, Pediatric Quality of Life Inventory, Pediatric Health Questionnaire 9, Revised Child Anxiety and Depression Scale–Short Version, Generalized Anxiety Disorder-7 item scale, Adolescent Sleep Wake Scale, Traumatic Brain Injury-Quality of Life Headache Pain score, and presence of suicidal ideation	Rehabilitation resulted in a –3.4-point change in HBI at 3 months (95% CI, –6.6 to –0.1) and 12 months (–4.1 points; 95% –7.7 to –0.4). PedsQL score was improved 4.7 points in intervention group (95% CI, 0.5 to 9.3 points) compared with control. Adolescent Sleep Wake Scale was improved by 0.4 (95% CI, 0.2 to 0.6; $p < .05$). No change in depressive, anxiety, or headache scores	Moderate

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Moore 2016	Prospective case series, with age stratification	14 adolescents/adults with persisting symptoms and disability after concussion Median age, 43 years; 43% male Source: New York University Langone Medical Center Concussion Program and Vestibular Rehabilitation Clinic	Supervised home exercise program consisting of vestibular rehabilitation combined with aerobic training for 6 months	Change in symptoms and disability: RPQ-3, RPQ-13, DHI, ABC, and FGA, and ability to RTW, RTA	Statistically significant improvements in symptoms and disability were observed between baseline and follow-up measurements at 3 months (RPQ-13 [$p < .002$]; DHI [$p < .001$]; and FGA [$p < .01$]) and at 6 months (RPQ-3 [$p < .001$]; RPQ-13 [$p < .004$]; DHI [$p < .001$]; ABC [$p < .007$]; FGA [$p < .001$])	High
O'Connor 2016	RCT	18 adult veterans with mTBI and mental illness Mean age, 51 years; 100% male Source: VA hospital	Vocational rehabilitation plus embedded cognitive rehabilitation vs. vocational rehabilitation only (control), for 12 weeks	Employment, including any work for pay, over a 1-year period	Numerical improvement in employment was observed with added cognitive rehabilitation vs. control over all time points, but differences were not statistically significant	High
Prangley 2017	Prospective case series	8 adolescents/adults with persisting symptoms after concussion (2 discontinued) Mean age, 41.63 years; % male not reported Source: physiotherapy clinic	Vestibular intervention (balance, vision, and neck strengthening exercises) customized to patient needs, for 4 weeks	Change in balance measures: REO, REC, SEO, SEC Nintendo Wii Board used to measure balance SCAT-3 used to assess symptoms	Statistically significant improvements in balance were observed across multiple testing conditions ($p < .04$) Objectively quantifying balance changes allows for a more sensitive measure of balance and detects changes unrecognizable to the naked eye	High
Reneker 2017	RCT	41 adolescents/adults with sports-related concussion and dizziness Mean age, ~16 years; 61% male Source: Walsh University and Akron Children's Hospital	Pragmatic, progressive physical therapy (intervention), individualized to patient needs vs. prescriptive, subtherapeutic treatment (control), for 4 weeks	Time to symptomatic recovery: Post-Concussion Scale Time to medical clearance for RTP	Time to medical clearance was significantly shorter with the intervention vs. control group (HR, 2.91; 95% CI, 1.01–8.43). Time to symptom recovery was faster in the intervention group (HR, 1.99; 95% CI, 0.95–4.15). Numerical but not statistically significant reduction in time to symptomatic recovery with intervention vs. control	Moderate
Rosner 2016	Retrospective case series	38 adolescents/adults with persisting symptoms and vertical heterophoria Mean age, 38.2 years; 34.2% male	Initial refractive and prism lenses (Prism Challenge) for 2–4 weeks	Change in subjective symptom burden: HDI, DHI, SAS, BVDQ, VAS	Compared with preintervention, binocular vision dysfunction was decreased 80.2% based on VAS ($p < .001$) and a relative reduction in the	High

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Schneider 2014	RCT	31 adolescents/adults with sport-related concussion and persisting symptoms Mean age, 15 years; ~44%–73% male, depending on group Source: University of Calgary Sport Medicine Centre	Vestibular rehabilitation with cervical spine physiotherapy vs. control (no vestibular rehabilitation/physiotherapy), for up to 8 weeks Both groups received postural education, range of motion exercises, and cognitive/physical rest until asymptomatic, then graded exertion	Time to medical clearance for RTP SCAT-2 used to assess symptoms	binocular vision dysfunction questionnaire of 50.5% ($p < .001$). Relative reduction in headache, dizziness, and anxiety scores A significantly higher rate of RTA by 8 weeks occurred in the intervention group vs. the control group (73% vs. 7%; 95% CI for difference 40% to 92.3%; $p < .001$). Those in the intervention group were 10.27 times (95% CI 1.51–69.56) more likely to RTA compared with control ($p < .001$)	Moderate
Teel 2021	RCT	62 adolescents with concussion Mean age 13.0–13.6 years; 42.9%–65% male Source: Montreal Children's Hospital ED	3D multiple object tracking protocol vs. video game control or standard of care alone	PCSI, presence of persistent postconcussive syndrome at 4 weeks	No significant impact on PCSI was found for child-reported ($p = .47$) and parent-reported ($p = .81$) symptoms. Persistent symptoms were seen in 43.8% of the treatment group and 33.3% in the control group ($p = .41$)	Moderate
Thastum 2018	Prospective case series	32 adolescents/adults with concussion and persisting symptoms Mean age, 23.3 years; 19% male Source: general practitioners in Central Denmark Region	“Get going after concussion,” an individually tailored, multidisciplinary, manualized intervention program, for up to 8 weeks	Change in symptoms, quality of life, and various physical and mental health scores: ESQ, RPQ, QOLIBRI-OS, B-IPQ, BRIQ, SCL-8	Statistically significant improvements RPQ (–8.9 points; 95% CI, –4.5 to –13.3; $p < .001$), QOLIBRI-OS (10.5 points; 95% CI, 2.5–18.5; $p = .010$), B-IPQ (–16.2 points; 95% CI, –9.7 to –22.7; $p < .001$), BRIQ “Limiting behavior” (–13.9 points; 95% CI, –3.2 to –24.6; $p = .011$), and BRIQ “All-or-nothing behavior” (–9.5 points; 95% CI, –0.7 to –18.3; $p = 0.035$). Postconcussive symptoms and quality of life scores improved and were maintained at 3 and 12 months ($p < .05$).	High

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Thiagarajan 2013 (this study and next 3 studies all part of same study and population)	Randomized, patient-blinded crossover, comparative	12 adults with chronic concussion (mTBI at least 1 year ago) with oculomotor symptoms Mean age, 29 years; 30% male Source: Raymond J. Greenwald Vision Rehabilitation Center	OMT vs. placebo (patients served as own controls) OMT performed before or after 6 weeks of placebo, with 1 week off between treatments	Change in clinical and laboratory vergence measures, visual attention, and near vision symptoms: NPC break and recovery, horizontal near phoria, horizontal near PFV and NFV ranges, vergence prism facility, and stereoacuity using the Titmus stereo test Vergence dynamics to symmetric step vergence stimuli VSAT, CISS	Compared with placebo, OMT resulted in statistically significant improvements in laboratory and clinical vergence parameters, visual attention, and near vision- related symptoms ($p < .05$)	Moderate
Thiagarajan 2014a	Randomized, patient-blinded crossover, comparative	12 adults with chronic concussion (mTBI at least 1 year ago) with oculomotor symptoms Mean age, 29 years; 30% male Source: Raymond J. Greenwald Vision Rehabilitation Center	OMT vs. placebo (patients served as own controls) OMT performed before or after 6 weeks of placebo, with 1 week off between treatments	Change in accommodative parameters (amplitude and facility), visual attention, and near vision symptoms: NPA, NRA, PRA, and accommodative facility dynamics VSAT, CISS	Compared with placebo, OMT resulted in statistically significant improvements in accommodative facility ($p = .01$; effect size = 1.1), accommodate amplitude ($p = .01$; effect size = 1.1), CISS score (37 to 28; $p < .01$), VSAT percentile (32nd to 50th; $p < .01$)	Moderate
Thiagarajan 2014b	Randomized, patient-blinded crossover, comparative	12 adults with chronic concussion (mTBI at least 1 year ago) with oculomotor symptoms Mean age, 29 years; 30% male Source: Raymond J. Greenwald Vision Rehabilitation Center	OMT vs. placebo (patients served as own controls) OMT performed before or after 6 weeks of placebo, with 1 week off between treatments	Improvement in reading rate, visual attention, and near vision symptoms: NPC, NPA using the push- up method, and reading eye movements Binocular horizontal versional eye movements Saccade ratio VSAT	Compared with placebo, OMT improved reading rate (142– 177 words per minute; $p < .01$), fixations (164 to 135 per 100 words; $p = .02$), and grade-level efficiency (4.1–6.3; $p = .01$). Mean saccade ratio reduced significantly from 2.1 to 1.7 ($p < .05$).	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Thiagarajan 2014c	Randomized, patient-blinded crossover, comparative	12 adults with chronic concussion (mTBI at least 1 year ago) with oculomotor symptoms Mean age, 29 years; 30% male Source: Raymond J. Greenwald Vision Rehabilitation Center	OMT vs. placebo (patients served as own controls) OMT performed before or after 6 weeks of placebo, with 1 week off between treatments	Change in binocular central fixation and saccadic gain, latency, and ratio	Compared with placebo, OMT significantly improved mean horizontal standard deviation for binocular central fixation ($p = .02$), saccadic gain in horizontal and vertical directions and for predictable stimulus horizontally, and mean saccade ratio for simulated reading multiple lines ($p = .0002$) but not for simulated reading single line	Moderate
Van Vleet 2015	Prospective comparative study	5 adults with chronic concussion and executive control deficits Mean age 41.4 years; 60% male Source: VA Northern California Health Care System	Tonic and phasic alertness training (TAPAT) vs. no training/weekly appointment reminders (control)	Change in attention and executive function: ACT, verbal fluency, Trails B, PCL-C or PCL-M task	TAPAT resulted in improved attention and executive function in 3 patients; improvements were not observed in controls	High
Vas 2016	RCT	60 community-dwelling adults (47 civilians, 13 veterans) with mild-spectrum TBI in chronic stages of recovery Mean age, ~41–42 years; ~52%–55% male, depending on group Source: flyers, website advertising, local brain injury support groups, representatives from local veteran centers	SMART versus BHW, for 8 weeks	Change in gist reasoning, TOSL	SMART led to statistically significant improvement in gist reasoning over time compared with BHW (4.19 vs. 3.72 pretraining; 5.16 vs. 4.25 posttraining; 5.13 vs. 3.32 3-month follow-up; $p = .02$)	Moderate
Yadav 2014	Prospective case series	7 adults with prolonged oculomotor deficits after concussion Mean age, 29.5 years; 14% male Source: Raymond J. Greenwald Rehabilitation Center/Brain Injury Clinic	OMT: included version, vergence, and accommodation, with an embedded and indirect visual attentional training component, for 6 weeks	Change in VEP and visual attention: alpha band (8–13 Hz) responsivity of VEP, VSAT	VEP amplitude and amplitude variability significantly decreased after the intervention ($p = .02$ for both comparisons). VEP latency and latency variability were not significantly changed, but these values were normal at baseline. VSAT percentile improved significantly ($p = .02$)	High

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Return to activity						
Darling 2014	Retrospective case series with prospective follow-up questionnaire	117 adolescents (91 interviewed) with sport-related concussion Mean age, 15.5 years; 75.2% male Source: University at Buffalo Sports Medicine Concussion Management Clinic	RTP decisions based on BCTT, computerized neuropsychological testing, and Zurich guidelines	Degree of success in RTP, ie, RTP with or without return of concussive symptoms Assessments: ANAM, ImPACT, SCAT-2	All athletes were successfully returned to sport after completing the BCTT and Zurich RTP protocol and continued to be engaged in play without exacerbation of symptoms at follow-up, according to self-report	High
Echlin 2010	Prospective case series	67 adolescent hockey players, 17 with concussion Mean age, 18.2 years; 100% male Source: Fourth-tier hockey players	Independent physician assessment and diagnosis of concussion, use of Zurich RTP protocol	Duration until RTP	15 of the 17 players with concussion achieved RTP during the study period; duration until RTP was noted to be higher than that reported in other studies	High
Kelleher 2014	Prospective case series	28 adolescent football players with mTBI Mean age 15.3 years; 100% male Source: Santa Barbara Cottage Hospital	RTP algorithm based on ACE, ImPACT, neurologic examination, and 1997 American Academy of Neurology guidelines	Mean time to RTP	The RTP algorithm used was successful in returning athletes in mean of 16.9 days	High
Ledoux 2022	Multicenter RCT	456 adolescents with concussion Mean age 13.3 years; 46.5%–43.7% male Source: 3 Canadian EDs	RTA initiation at 72 h after injury vs. delayed RTA protocol initiation when symptom free	HBI, BESS, adverse events	No significant difference in HBI was noted on intent-to-treat analysis (mean difference = −1.3; 95% CI: −3.6 to 1.1; $p = .30$). Per protocol analysis revealed significant change in HBI score with intervention vs. control (mean difference = −4.3; 95% CI, −8.4 to −0.2; $p = .038$)	Moderate
Philipson 2021	Retrospective cohort study	115 adolescents with concussion High school age; proportion male not stated Source: 5 Seattle-area high schools	Academic accommodations between participants requiring less than 2 weeks vs. 3 to 4 weeks on RTA protocol were compared	SCAT-5, types of accommodations, number of accommodations per week	Participants in the long duration group reported more unique symptoms (mean 13.6 vs. 11.6; $p = .038$), higher total severity score (mean 40.5 vs. 29.7; $p = .005$), and higher average severity per symptom (mean 2.9 vs. 2.4; $p = .007$)	Moderate

(Continues)

TABLE 2 (Continued)

Author/year	Study design	Population/source	Intervention/exposure	Outcome measures/tools	Results	Risk of bias
Wiebe 2022	Prospective cohort study	1715 adolescent athletes participating in contact sports Mean age 20 years; 68.4% male Source: 20 college athletic programs	Site-based RTP protocol	RTP within 14 days, return to sport >28 days	Athletes who engaged in activity after symptom resolution returned to sport sooner than those that returned to academics early (11 days vs. 14 days; $p < .001$). Return to activity and learning on the same day as symptom resolution was associated with faster RTP (median 10 days vs. 14 days; $p < .001$ for activity and 11 days vs. 16 days; $p < .001$ for learning).	High

Abbreviations: ABC, Activities-specific Balance Confidence; ACE, Acute Concussion Evaluation; ACT, auditory consonant trigrams; ANAM, Automated Neuropsychological Assessment Metrics; BCTT, Buffalo Concussion Treadmill Test; BESS, Balance Error Scoring System; BHW, Brain Health Workshop; B-IPQ, Brief Illness Perception Questionnaire; Borg CR10 RPE, Borg Category Ratio 10 Rating of Perceived Exertion; BRIQ, Behavioral Response to Illness Questionnaire; BVDQ, Binocular Vision Dysfunction Questionnaire; CI, confidence interval; CISS, Convergence Insufficiency Symptom Survey; CNT, computerized neuropsychological tests; DGI, Dynamic Gait Index; DHI, Dizziness Handicap Inventory; D-KEFS, Delis-Kaplan Executive Function System; ED, emergency department; ESQ, Experience of Service Questionnaire; FGA, Functional Gait Assessment; fMRI, functional magnetic resonance imaging; FSIQ-2, Full Scale Intelligent Quotient-2; FTSTS, Functional Gait Assessment Timed Up and Go; GSC, Graded Symptom Checklist; HADS, Hospital Anxiety and Depression Scale; HBI, Health and Behavior Inventory; HDI, Headache Disability Index; HIMAT, High-Level Mobility Assessment Tool; HR, hazard ratio; ImPACT, Immediate Post-Concussion Assessment and Cognitive Testing; IQR, interquartile range; KBCI, Key Behaviors Change Inventory; LiSat-11, Life Satisfaction Questionnaire; MPAL-4, Mayo-Portland Adaptability Inventory-4; mTBI, mild traumatic brain injury; NFV, negative fusional vergence; NPA, near point of accommodation; NPC, near point of convergence; OMT, oculomotor therapy; OR, odds ratio; PASAT, Paced Auditory Serial Addition Test; PCL-C, Post-Traumatic Stress Disorder Checklist, civilian; PCL-M, Post-Traumatic Stress Disorder Checklist, military; PCSJ, Post-Concussion Symptom Inventory; PCSQ, Post-Concussion Symptoms Questionnaire; PCSS, Post-Concussion Symptom Scale; PedsQL, Pediatric Quality of Life Inventory; PFV, positive fusional vergence; PPCSJ, Persistent Postconcussive Symptoms; PTSQ, posttraumatic stress disorder; QOLIBRI, Quality of Life After Brain Injury; QOLIBRI-OS, Quality of Life After Brain Injury-Overall Scale; RCT, randomized controlled trial; REC, Romberg eyes closed; REO, Romberg eyes open; RPQ-13, Rivermead Post-Concussion Questionnaire, function; RPQ-3, Rivermead Post-Concussion Questionnaire, symptom; RTA, return to activity; RTP, return to play; RTW, return to work/study; SAC, Standard Assessment of Concussion; SAS, Zung Self-Rating Anxiety Scale; SCAT-2, Sport Concussion Assessment Tool-2; SCAT-3, Sport Concussion Assessment Tool-3; SCAT-5, Sport Concussion Assessment Tool-5; SCL-8, Symptom Checklist 8; SCL-90-R, Symptom Checklist-19 Revised; SDMT, Symbol Digit Modalities Test; SEC, single leg eyes closed; SEO, single leg eyes open; SF-12, Short Form Survey 12; SMART, Strategic Memory Advanced Reasoning Training; SMS, Service Members; SOT, Sensory Organization Test; SSTEP, Sub-Symptom Threshold Exercise Program; TOSL, Test of Strategic Learning; VA, Veterans Affairs; VAS, visual analogue scale; VEP, visually evoked potentials; VIS, Vocational Independence Scale; VOMS, visual/ocular motor screen; VSAT, Visual Search and Attention Test; VSS-A, Vertigo Symptom Scale-Automatic Anxiety Symptoms; VSS-SF, Vertigo Symptom Scale-Short Form; VSS-V, Vertigo Symptom Scale-Vertigo Balance.

were categorized by intervention subtype, population age group, and diagnosis, whether consistent with acute or chronic concussion. Intervention subtypes included cognitive, physical, and combined cognitive and physical rest; exercise as a single category of all identified studies involved some form of aerobic exercise; cognitive, oculomotor, vestibular, and multimodal rehabilitation; and RTA protocol as a single category of all identified studies involved use of standard RTA protocols (eg, Zurich return to play [RTP] protocol¹) or similar algorithms. See the [Appendix](#) for full definitions of the interventions.

For the purpose of this review, population age was categorized as children (5–11 years as there are not validated tools or sufficient literature for those under 5), adolescents (12–21 years), or adults (>21 years) according to age groups recommended for clinical trial design,³⁵ which are based on the Eunice Kennedy Shriver National Institute of Child Health and Human Development Pediatric Terminology and recommendations from the American Academy of Pediatrics, Centers for Disease Control and Prevention, and other relevant organizations. If the population age range was not reported, the mean age was used to categorize the study population. Diagnosis was categorized as acute or chronic concussion based on the diagnostic description/definition provided by study authors. Acute concussion included injuries described as occurring within 1 week from initial physician assessment, and/or any other terminology suggestive of a recent-onset condition. Chronic concussion included postinjury conditions described as postconcussion syndrome, prolonged or persisting symptoms, prolonged or slow recovery, late sequelae and/or any other terminology suggestive of an enduring condition.

Panel consensus process

A modified version of the RAND/University of California Los Angeles Appropriateness Method was used to develop the panel consensus on interventions for concussion management.³⁶ Panel members were paired to serve as primary reviewers of the available evidence for each clinical question. Each member pair reviewed their assigned evidence and then presented the findings to the panel at an in-person meeting in February 2019. Due to time constraints and the COVID-19 pandemic, subsequent meetings occurred online with Zoom web conferencing software. A final online conference was held in September 2022 to assess evidence derived from the updated literature search. Reviewers were assigned specific ownership of certain literature in pairs. These pairs presented the literature in detail to the full panel for discussion. The panel then discussed discrepancies in the data extraction and quality analysis until a consensus was reached. After the initial

presentation of evidence, panel members voted independently on the appropriateness of the interventions (benefits outweigh harms) by ranking appropriateness on a scale of 1 to 9 (1 = highly inappropriate; 9 = highly appropriate), based on an assessment of the relative benefits and harms identified in the available evidence. The RAND/University of California Los Angeles Appropriateness Method defines an appropriate intervention as one for which “the expected health benefit...exceeds the expected negative consequences...by a sufficiently wide margin that the procedure is worth doing, exclusive of cost;³⁶ this definition of appropriate was used in this evaluation.” Median scores and mean deviations were calculated, which corresponded to levels of appropriateness (score 1–3 = inappropriate; 4–6 = uncertain; and 7–9 = appropriate) and status of consensus (agreement = ≤ 3 responses outside 3-point boundary of median; disagreement = ≥ 4 responses outside 3-point boundary of median). After reviewing the initial voting results, the panel discussed the evidence in detail to clarify any questions that panel members had regarding the evidence and any discrepancies in voting results, with the goal of achieving a consensus on the evidence. Panel members voted independently a second time, which provided the final voting results supporting the consensus (Table 3).

RESULTS

A total of 6303 articles were retrieved through the systematic literature search and screened for inclusion. After duplicates and studies not meeting inclusion criteria were removed, 81 unique original studies (8 of which contributed to more than one PICO question) were reviewed in full and included in this review and consensus process. Results of the literature search are depicted in PRISMA flow charts in the [Appendix](#) and summarized herein for each PICO question.

Overarching PICO: In patients with concussion, does rest versus no/other intervention improve outcomes?

General clinical question: Does rest improve outcomes for patients with concussion?

Studies addressing rest were subdivided into three categories based on rest type—cognitive, physical, and combined cognitive and physical rest.

Cognitive rest

Four studies investigating the role of cognitive rest in children, adolescents, and/or adults with acute concussion were identified, including one RCT,³⁷ two prospective

TABLE 3 Final voting results and panel consensus on interventions for concussion.

	Acute or chronic concussion	Age group	Evidence	Panel median	Mean deviation	Appropriateness of intervention	Consensus status
Rest							
Cognitive rest	Acute	Children	Brown 2014 Brooks 2016 Gibson 2013	5	0.29	Uncertain	Agreement
		Adolescents	Brown 2014 Brooks 2016 Gibson 2013 Varner 2017	5	0.49	Uncertain	Agreement
		Adults	Brown 2014 Gibson 2013 Varner 2017	5	0.57	Uncertain	Agreement
Physical rest	Acute	Children	Grool 2016 Howell 2016	4	0.94	Uncertain	Agreement
		Adolescents	Grool 2016 Howell 2016 Maerlender 2015 Willer 2019	4	0.56	Uncertain	Agreement
Combined cognitive/ physical rest	Acute	Adults	Howell 2016	4	0.90	Uncertain	Agreement
		Children	Root 2020	5	0.72	Uncertain	Agreement
		Adolescents	Buckley 2016 de Kruijk 2002 Moser 2012 Sufrinko 2017 Taubman 2016 Thomas 2015 Root 2020	5	0.98	Uncertain	Agreement
	Adults	de Kruijk 2002 Moser 2012	5	0.69	Uncertain	Agreement	
	Chronic	Adolescents	Moser 2015	5	0.49	Uncertain	Agreement
Exercise							
Exercise	Acute	Children	Grool 2016 Howell 2016 Majerske 2008	5	0.66	Uncertain	Agreement
		Adolescents	Buckley 2022 Grool 2016 Howell 2016 Howell 2020 Howell 2021 Howell 2022 Lawrence 2018 Leddy 2019 Leddy, Haider and Ellis 2019 Leddy 2021 Maerlender 2015 Majerske 2008 Popovich 2021 Seehusen 2021 Stumph 2020 Worts 2022 Yang 2021	8	0.50	Appropriate	Agreement
	Adults	Howell 2016 Remigio-Baker 2020 Varner 2021	6	0.75	Uncertain	Agreement	
	Chronic	Children	Chrisman 2017 Cordingley 2016 Dobney 2017	6	0.75	Uncertain	Agreement
		Adolescents	Chan 2018	6	0.44	Uncertain	Agreement

TABLE 3 (Continued)

Intervention	Acute or chronic concussion	Age group	Evidence	Panel median	Mean deviation	Appropriateness of intervention	Consensus status
			Chrisman 2017 Cordingley 2016 Coslick 2020 Dobney 2017 Gagnon 2016 Grabowski 2017 Kurowski 2017 Leddy 2010 Leddy 2013 Moore 2016 Snyder 2021				
		Adults	Leddy 2013 Leddy 2010 Larson-Dupuis 2021 Moore 2016 Snyder 2021	6	0.75	Uncertain	Agreement
Rehabilitation							
Cognitive rehabilitation	Acute	Adolescents	Teel 2021	5	0.56	Uncertain	Agreement
	Chronic	Adults	Cooper 2017 Han 2018 Mahncke 2021 O'Connor 2016 Van Vleet 2015 Vas 2106	6	0.44	Uncertain	Agreement
Oculomotor rehabilitation	Chronic	Adolescents	Ciuffreda 2008 Rosner 2016	5	0.35	Uncertain	Agreement
		Adults	Ciuffreda 2008 Rosner 2016 Thiagajagan 2013 Thiagajagan 2014-Effect Thiagajagan 2014-oculomotor Thiagajagan 2014-versional Yadav 2014	6	0.59	Uncertain	Agreement
Vestibular rehabilitation	Acute	Adolescents	Kontos 2021	6	0.75	Uncertain	Agreement
	Chronic	Children	Alsalaheen 2010	6	0.49	Uncertain	Agreement
		Adolescents	Ahluwalia 2021 Alsalaheen 2010 Alsalaheen 2020 Pranglely 2017 Schneider 2014	6	0.47	Uncertain	Agreement
		Adults	Alsalaheen 2010 Klefffelgaard 2016 Pranglely 2017 Schneider 2014	6	0.35	Uncertain	Agreement
Multimodal rehabilitation	Acute	Adolescents	Elgmark 2007 Reneker 2017	6	0.62	Uncertain	Agreement
		Adults	Dharm-Datta 2015 Elgmark 2007 Reneker 2017	5	0.59	Uncertain	Agreement
	Chronic	Adolescents	Chan 2018 Dobney 2017 Gagnon 2016 Gauvin-Lepage 2020 Grabowski 2017 McCarty 2021 Moore 2016 Thastum 2018	6	0.38	Uncertain	Agreement

(Continues)

TABLE 3 (Continued)

Intervention	Acute or chronic concussion	Age group	Evidence	Panel median	Mean deviation	Appropriateness of intervention	Consensus status
		Adults	Hammerle 2019 Langevin 2022 Moore 2016 Thastum 2018	6	0.22	Uncertain	Agreement
Return to activity (RTA)							
RTA protocol	Acute	Adolescents	Darling 2014 Echlin 2010 Kelleher 2014 Ledoux 2022 Phillipson 2021 Wiebe 2022	6	0.69	Uncertain	Agreement

comparative studies,^{38,39} and one retrospective comparative study.⁴⁰ Although a single nonrandomized study associated lower levels of cognitive activity with shorter symptom duration,³⁸ the remaining three studies did not find any benefit. These studies were of moderate to high risk of bias and were of insufficient quality to recommend for or against cognitive rest as a therapy for acute concussion in a given age group. No evidence was found addressing the effect of cognitive rest on outcomes of chronic concussion. Based on the available evidence, the panel voted on the appropriateness of cognitive rest for the management of children, adolescents, and adults with acute concussion, resulting in the consensus that the panel is uncertain whether cognitive rest is appropriate for these age groups (Table 3).

Physical rest

All evidence identified on physical rest for concussion involved children, adolescents, and/or adults with acute concussion. Four relevant studies were retrieved: one RCT and three prospective comparative studies.^{41–44} Notably, in many of these studies, physical rest was not the intervention of interest but was the role of the control group (versus physical activity) or a study group categorized by low levels of physical activity. One study compared a cohort prescribed rest to historical cohorts prescribed exercise or an active control.⁴⁴ All prospective comparative studies suggested physical activity was associated with improved outcomes, and the randomized trial had a negative result. No evidence was found on the effects of physical rest for chronic concussion. Based on the available evidence, the panel voted on the appropriateness of physical rest for the management of children, adolescents, and adults with acute concussion, resulting in the consensus that the panel is uncertain whether physical rest is appropriate for these age groups (Table 3).

Combined cognitive/physical rest

The evidence for combined cognitive/physical rest included studies in patients with acute and chronic concussion, discussed separately.

Acute concussion

For acute concussion, the evidence comprises seven studies: two RCTs of which one underwent secondary analysis,^{45–47} two prospective comparative studies,^{48,49} one retrospective comparative study,⁵⁰ and one retrospective case series⁵¹ that involved children, adolescents, and/or adults. Studies of less robust designs did suggest improvement in outcomes with combined physical/cognitive rest; however, the randomized trials demonstrated either a negative result or a detrimental effect of combined cognitive/physical rest. Evidence is mixed as to whether combined cognitive and physical rest improves outcomes for adolescent and adult patients with acute concussion, with studies variously supporting beneficial effect, detrimental impact, or no effect on outcomes. The studies on cognitive and physical rest for acute concussion were determined to have a moderate to high risk of bias. Based on the available evidence, the panel is uncertain whether combined cognitive and physical rest is appropriate for acute concussion (Table 3).

Chronic concussion

The evidence on combined cognitive/physical exercise for chronic concussion was limited, with only one retrospective case series (involving adolescents) identified.⁵² This study was not of sufficient quality to determine if combined cognitive/physical rest is an appropriate therapy for adolescence with chronic concussion. Based on the available evidence, the panel is uncertain whether combined cognitive and physical rest is appropriate for this age group (Table 3).

Exercise

Overarching PICO: In patients with concussion, does exercise therapy versus no/other intervention improve outcomes?

General clinical question: Does exercise improve outcomes for patients with concussion?

The evidence on exercise included studies in patients with acute or chronic concussion, discussed separately.

Acute concussion

For acute concussion, evidence concerning exercise included 20 studies: 7 RCTs,^{41,53–58} 8 prospective comparative studies,^{42,43,59–64} and 5 retrospective comparative studies^{50,65–68} in children, adolescents, and/or adults. Most studies involved adolescents. Two of the RCTs involving adolescents were of sufficiently high quality and low to moderate risk of bias to conclude that aerobic exercise is an appropriate therapy for adolescents with acute concussion.^{53,55} Another group of studies — with lower overall quality — reported a range of outcomes ranging from beneficial to mixed to negative. The remaining studies involving adolescents and studies involving other age groups were not of sufficient quality to recommend for or against exercise as a therapy for acute concussion in any other age group. Based on the available evidence, the panel voted on the appropriateness of exercise for the management of children, adolescents, and adults with acute concussion, resulting in the consensus that exercise is an appropriate intervention for adolescents with acute concussion, although the panel is uncertain whether there is evidence that exercise is appropriate for other age groups (Table 3).

Chronic concussion

The evidence on exercise for chronic concussion involved children, adolescents, and/or adults and included 13 studies: 3 RCTs,^{69–71} 2 prospective comparative studies,^{72,73} 3 prospective case series,^{74–76} 1 retrospective comparative study,⁷⁷ and 4 retrospective case series.^{78–81} Most studies of lower level of evidence suggested a beneficial effect of exercise on chronic concussion. The evidence is inconclusive as to whether exercise improves outcomes for patients with chronic concussion. Many studies showed some benefit of the interventions evaluated, but most studies lacked control groups and evaluated exercise as part of multimodal strategies, which limited the ability to confirm the effect of exercise alone on recovery. The available randomized studies had conflicting positive and negative results. The evidence was determined to include a range of moderate to high risk of bias. Based on the available evidence, the panel voted on the appropriateness of exercise for the management of children, adolescents, and adults with chronic concussion, resulting in

the consensus that the panel is uncertain whether exercise is appropriate for these age groups.

Rehabilitation

Overarching PICO: In patients with concussion, does rehabilitation versus no/other intervention improve outcomes?

General clinical question: Does rehabilitation improve outcomes for patients with concussion?

Studies addressing rehabilitation were subdivided into four categories based on rehabilitation type including cognitive, oculomotor, vestibular, and multimodal.

Cognitive rehabilitation

Seven studies, including six RCTs and one small prospective comparative study,^{82–88} were identified on cognitive rehabilitation. One study involved adolescents with acute concussion. All other studies involved adults with chronic concussion. No evidence was found for the effect of cognitive rehabilitation in children or adults with acute concussion or for children or adolescents with chronic concussion. Evidence is mixed as to whether cognitive rehabilitation improves outcomes for patients with chronic concussion, with five RCTs reporting some benefit in cognitive function after their respective interventions, one study showing no statistically significant benefit, and the final study being too small to confirm any benefits from the cognitive rehabilitation intervention. The evidence evaluated was insufficient to recommend for or against cognitive rehabilitation for concussion in any group. Based on the available evidence, the panel voted on the appropriateness of cognitive rehabilitation for the management of adolescents and adults with concussion, resulting in the consensus that the panel is uncertain whether cognitive rehabilitation is appropriate for these age groups.

Oculomotor rehabilitation

The evidence on oculomotor rehabilitation comprised seven studies: one randomized crossover study with four separate analyses and three case series (one prospective and two retrospective), all in adolescents and/or adults with chronic concussion.^{89–95} No evidence was found in any age group with acute concussion or in children with chronic concussion. The evidence on oculomotor rehabilitation shows some benefit with respect to improving prolonged oculomotor deficits in patients with chronic concussion, but due to heterogeneity in specific outcomes measured in each study, lack of control groups in the case series, and no subgroup analysis by age, it was difficult for the panel to draw any firm conclusions about efficacy of the interventions. The studies were determined to have moderate to high risk of bias. Based on the available

evidence, the panel voted on the appropriateness of oculomotor rehabilitation for the management of adolescents and adults with chronic concussion, resulting in the consensus that the panel is uncertain whether oculomotor rehabilitation is appropriate for these age groups.

Vestibular rehabilitation

The evidence on vestibular rehabilitation comprised seven studies: 2 RCTs,^{96,97} one mixed comparative study,⁹⁸ one retrospective comparative study,⁹⁹ and three case series (two prospective and one retrospective)^{100–102} in children with chronic concussion, adolescents with acute and chronic concussion, and adults with chronic concussion. No evidence was found for children or adults with acute concussion. Due to differences in outcomes measured across studies and design-related limitations of most studies, the evidence was determined to be of moderate to high risk of bias; thus, the benefits observed were difficult to confirm. Based on the available evidence, the panel voted on the appropriateness of vestibular rehabilitation for the management of adolescents with acute concussion and children, adolescents, and adults with chronic concussion, resulting in the consensus that the panel is uncertain whether the intervention is appropriate for these age groups.

Multimodal rehabilitation

The evidence on multimodal rehabilitation included studies in acute and chronic concussion, discussed separately.

Acute concussion. In acute concussion, the evidence comprised three studies: two RCTs and one prospective case series in adolescent and/or adult patients.^{103–105} No evidence was found in children with acute concussion. The evidence on multimodal rehabilitation for adolescent and adult patients with acute concussion showed no significant effect on symptom improvement, but one RCT found a significantly shorter time to medical clearance for return to play with rehabilitation, and another RCT found improvements in physical health scores with rehabilitation. The studies evaluated for acute concussion were determined to have moderate to high risk of bias. Based on the available evidence, the panel voted on the appropriateness of multimodal rehabilitation for the management of adolescents and adults with acute concussion, resulting in the consensus that the panel is uncertain whether this intervention is appropriate for these age groups (Table 3).

Chronic concussion. The evidence on multimodal rehabilitation for chronic concussion includes 10 studies: 3 RCTs,^{69,106,107} 1 prospective comparative study,¹⁰⁸ 1 retrospective comparative study,¹⁰⁹ and 5 case series (3 prospective and 2 retrospective)^{75,76,80,81,110} in adolescents and/or adults. No evidence was found on multimodal rehabilitation for children with chronic concussion.

All evidence on multimodal rehabilitation for adolescents and adults with chronic concussion showed statistically significant improvements in symptoms and other measures; however, the RCTs had moderate risk of bias, and the other studies were case series with no control groups and other limitations that introduced a high risk of bias. Based on the available evidence, the panel voted on the appropriateness of multimodal rehabilitation for the management of adolescents and adults with acute concussion and chronic concussion, resulting in the consensus that the panel is uncertain whether this intervention is appropriate for these age groups (Table 3).

RTA protocol

Overarching PICO: In patients with concussion, do RTA protocols versus no/other intervention improve outcomes?

General clinical question: Do RTA protocols improve outcomes for patients with concussion?

The evidence identified on RTA protocols comprised six studies: one RCT,¹¹¹ one prospective cohort study,¹¹² one retrospective cohort study,¹¹³ two prospective case series,^{23,114} and one retrospective case series.¹¹⁵ The studies involved RTA protocols in adolescents with acute concussion. Findings from many lower quality studies suggest that RTA protocols may be safe and effective for returning patients to sport, but the noncomparative designs and other study limitations that introduced a high risk of bias prevented the panel from determining the actual benefits of these protocols in terms of post-concussion outcomes. The available RCT did not demonstrate benefit from an RTA protocol. Based on the available evidence, the panel voted on the appropriateness of RTA protocols for the management of adolescents with acute concussion, resulting in the consensus that the panel is uncertain whether RTA protocols are appropriate for use in this patient population (Table 3).

DISCUSSION

The panel reviewed multiple studies for the clinical questions on rest, exercise, rehabilitation, and RTA protocols. The panel noted a benefit of aerobic exercise for adolescents with acute concussion. Although treatment effects for other therapies and other age groups were observed in some of the studies, suggesting that these therapies may have some benefit for some patients, limitations in study design and lack of subgroup analyses in many cases precluded the panel from drawing definitive conclusions about the appropriateness of these therapies for different populations of concussion. Clinicians should be aware that the

appropriateness of exercise as a therapy for concussion in terms of symptomatic improvement does not inform when RTA is appropriate. The panel found evidence of detrimental effects from strict rest and high-intensity physical activity. No other interventions were associated with adverse outcomes, which suggests that most interventions are safe to implement, if deemed reasonable in a given clinical scenario while considering individual patient needs, values, and preferences.

Importantly, results of this review must be interpreted with caution due to the limitations and risk of bias of each study and of the collective body of evidence for each clinical question. Most of the studies were determined to be of moderate to high risk of bias. Although randomized trials were included in this review, many of them lacked rigorous randomization methods, allocation concealment, blinding, and power calculations and, therefore, only moderate-quality evidence. Almost all studies reviewed had the potential for selection bias, because patients were mainly recruited from specialty concussion clinics or hospital emergency departments, suggesting that these patients sought medical care because they likely had more severe symptoms than the general concussion population. It could be argued that patients without symptoms may not need therapy and would not be the focus of concussion management studies, but some evidence suggests that even patients with no symptoms can still have cognitive and functional deficits according to objective measures.²³ Treatment recommendations based on study findings that are not generalizable to asymptomatic patients may lead to suboptimal care of asymptomatic patients with residual cognitive and functional impairments not easily diagnosed by clinical evaluation alone. Future studies must seek recruitment from a broader range of sources—schools, after school sports complexes, and primary care clinics—in addition to specialty clinics and hospitals—to ensure adequate representation of all populations with concussion. Consideration must also be given to the influence of health disparities on access to care for individuals sustaining a concussion.

In addition, appropriate subanalyses must be incorporated into the design of future studies to assess the independent effect of interventions on different populations. Although some of the reviewed studies did incorporate subanalyses to evaluate differences in outcomes by postinjury symptom scores, age, and other factors, many of these studies had only one treatment group. Lack of a comparator group resulted in difficulty determining if patients recovered because of the intervention or because of spontaneous recovery. The possibility of spontaneous recovery remains problematic even in comparative studies, but adequate study designs may help to minimize this confounding factor. For example, one study included a 1-week lead-

in period between the initial clinic visit and the start of the intervention to monitor for any apparent recovery in the treatment groups.

The potential for measurement bias was also moderate to high in many of the reviewed studies. Although most studies incorporated the use of validated tools for symptom reporting, these tools have limitations and are based on patient self-report, which increases the risk of recall bias and under- or overreporting of symptoms. Underreporting may lead to premature RTA and, consequently, increased risk of persisting symptoms and future concussion. Alternatively, overreporting may lead to prolonged rest and removal from activity, which may have negative implications for recovery and mental and physical wellbeing.^{46,116,117} Few studies in this review incorporated objective tools for evaluating other measures of concussion severity and recovery (eg, neurocognitive and physical tests). Additionally, adherence to treatment groups was typically measured by patient self-report. Tools, such as accelerometers, pedometers, and activity tracking devices, were used in some cases, but should be incorporated into more studies to measure activity levels and adherence to assigned groups.

Other limitations found in some of the studies were the lack of strict definitions to confirm the diagnosis of concussion, small and variable patient populations, variable timeframes of intervention initiation and duration, lack of control groups and randomization, retrospective designs, and lack of subgroup analyses and multivariate analyses to determine independent effects of interventions on different patient populations. Future research should aim to address these limitations. The general lack of high-quality studies highlights the difficulties in studying the concussion population. Specific challenges relate to enrolling representative populations, adequately controlling conditions, completing data collection, and objectively assessing outcomes; these factors present barriers to rigorous evaluation of therapeutic interventions for concussion. Additionally, the lack of an objective clinical standard for defining and diagnosing concussion limits the ability to generate a standardized clinical practice guideline for management of concussion.

The panel consensus on each intervention must be considered in the context of several limitations of this review. The systematic literature search was confined to a single database (PubMed) and a 20-year publication range; therefore, relevant evidence published elsewhere or outside of this data range was not included. However, PubMed is a comprehensive database including “more than 34 million citations for biomedical literature from MEDLINE, life science journals, and online books,”¹¹⁸ and a 20-year range in publication date would be expected to be representative of the most current practices in concussion management. Evidence may have been excluded inadvertently for the

intervention subcategories because the systematic searches were performed for the overarching PICO questions only. After the interventions were subcategorized, no attempt was made to expand the literature search to include all relevant terms for those subcategories. The inclusion and exclusion criteria did not include a strict definition for concussion; any study describing patients as having a concussion was included. Because definitions varied across studies—and in some cases were not provided at all—patients without confirmed concussion may have been included in this analysis. Further, definitions for specific postconcussion outcomes were not required for inclusion in this review. All relevant outcomes were included irrespective of whether they were well defined in the individual studies, although most of the studies used validated patient-reported outcome measurement tools. The heterogeneity in study populations, interventions, and outcomes precluded a formal meta-analysis of data on each overarching PICO question. This systematic review was also performed according to the 2009 PRISMA statement,³¹ rather than the 2020 update that was released after the initial portion of this review had largely been completed¹¹⁹; thus, some aspects of the review process may have been modified if the current methods had been incorporated. Finally, due to resource limitations related to the COVID-19 pandemic, the review of evidence and consensus process took more than 5 years to complete. Logistical, professional, and personal challenges related to the COVID-19 pandemic may have introduced bias (decision/respondent fatigue, recall bias) and inconsistency in the review and consensus process.

CONCLUSIONS

After a systemic review of literature, the authors concluded that subsymptom threshold aerobic exercise is an appropriate intervention, contributing to symptomatic improvement, for adolescents with acute concussion/mTBI. The evidence for exercise in other age groups and for other therapeutic interventions for concussion remains limited to small RCTs and observational studies of moderate to low quality. Although the panel found no strong evidence to support or recommend against the additional interventions evaluated in this review, most interventions appear to be safe if used judiciously for appropriate patients. Of note, the recommendation for exercise in adolescents with acute concussion does not determine when clearance for RTA is appropriate. These findings may help to inform future RCTs and systematic reviews on rest, exercise, rehabilitation, and RTA protocols in acute and chronic concussion. Clinicians should remain aware of new evidence that becomes available and use best clinical judgment in initiating any of the evaluated interventions for patients

with concussion, while considering the best available evidence at the time and each patient's needs, values, and preferences.

ACKNOWLEDGMENTS

The panel acknowledges AAPM&R staff members Melanie Dolak for reviewing and providing feedback on this manuscript and Megan Roop for providing editing assistance.

DISCLOSURE

Jennifer Swanson is a medical research and writing consultant paid by AAPM&R. Authors have no financial interests to disclose in relation to this project.

ORCID

Arthur J. De Luigi  <https://orcid.org/0000-0003-4610-529X>

Kathleen R. Bell  <https://orcid.org/0000-0002-0928-2046>

REFERENCES

1. McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport—the 5(th) international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med*. 2017;51(11):838-847. doi:10.1136/bjsports-2017-097699
2. McCrory P, Feddermann-Demont N, Dvorak J, et al. What is the definition of sports-related concussion: a systematic review. *Br J Sports Med*. 2017;51(11):877-887. doi:10.1136/bjsports-2016-097393
3. Halstead ME, Walter KD, Moffatt K, Council On Sports M, Fitness. Sport-related concussion in children and adolescents. *Pediatrics*. 2018;142(6):1-24. doi:10.1542/peds.2018-3074
4. McInnes K, Friesen CL, MacKenzie DE, Westwood DA, Boe SG. Mild traumatic brain injury (mTBI) and chronic cognitive impairment: a scoping review. *PLoS One*. 2017;12(4):e0174847. doi:10.1371/journal.pone.0174847
5. Neelakantan M, Ryali B, Cabral MD, Harris A, McCarroll J, Patel DR. Academic performance following sport-related concussions in children and adolescents: a scoping review. *Int J Environ Res Public Health*. 2020;17(20):7373. doi:10.3390/ijerph17207602
6. Harmon KG, Drezner JA, Gammons M, et al. American Medical Society for Sports Medicine position statement: concussion in sport. *Br J Sports Med*. 2013;47(1):15-26. doi:10.1136/bjsports-2012-091941
7. Carroll EL, Outtrim JG, Forsyth F, et al. Mild traumatic brain injury recovery: a growth curve modelling analysis over 2 years. *J Neurol*. 2020;267(11):3223-3234. doi:10.1007/s00415-020-09979-x
8. Laker SR. Epidemiology of concussion and mild traumatic brain injury. *PM R*. 2011;3(10 Suppl 2):S354-S358. doi:10.1016/j.pmrj.2011.07.017
9. Stein MB, Jain S, Giacino JT, et al. Risk of posttraumatic stress disorder and major depression in civilian patients after mild traumatic brain injury: a TRACK-TBI study. *JAMA Psychiatry*. 2019;76(3):249-258. doi:10.1001/jamapsychiatry.2018.4288
10. Finkbeiner NW, Max JE, Longman S, Debert C. Knowing what we don't know: long-term psychiatric outcomes following adult concussion in sports. *Can J Psychiatry*. 2016;61(5):270-276. doi:10.1177/0706743716644953
11. Patricios JS, Arden CL, Hislop MD, et al. Implementation of the 2017 Berlin concussion in sport group consensus statement in contact and collision sports: a joint position statement from

- 11 national and international sports organisations. *Br J Sports Med.* 2018;52(10):635-641. doi:[10.1136/bjsports-2018-099079](https://doi.org/10.1136/bjsports-2018-099079)
12. Cassidy JD, Carroll LJ, Peloso PM, et al. Incidence, risk factors and prevention of mild traumatic brain injury: results of the WHO collaborating Centre task force on mild traumatic brain injury. *J Rehabil Med.* 2004;36(43 Suppl):28-60. doi:[10.1080/16501960410023732](https://doi.org/10.1080/16501960410023732)
13. Cameron KL, Marshall SW, Sturdivant RX, Lincoln AE. Trends in the incidence of physician-diagnosed mild traumatic brain injury among active duty U.S. military personnel between 1997 and 2007. *J Neurotrauma.* 2012;29(7):1313-1321. doi:[10.1089/neu.2011.2168](https://doi.org/10.1089/neu.2011.2168)
14. Wagner R, Zach K, Kobayashi Y, Gottschalk AW. Race and concussion: an emerging relationship. *Ochsner J Winter.* 2020;20(4):348-349. doi:[10.31486/toj.20.0145](https://doi.org/10.31486/toj.20.0145)
15. Lyons TW, Miller KA, Miller AF, Mannix R. Racial and ethnic differences in emergency department utilization and diagnosis for sports-related head injuries. *Front Neurol.* 2019;10:690. doi:[10.3389/fneur.2019.00690](https://doi.org/10.3389/fneur.2019.00690)
16. Wallace J, Covassin T, Moran R. Racial disparities in concussion knowledge and symptom recognition in American adolescent athletes. *J Racial Ethn Health Disparities.* 2018;5(1):221-228. doi:[10.1007/s40615-017-0361-1](https://doi.org/10.1007/s40615-017-0361-1)
17. Campbell J, Howland J, Hess C, et al. Disparities in baseline neurocognitive testing for student concussion management in Massachusetts high schools. *BMJ Open Sport Exerc Med.* 2020;6(1):e000752. doi:[10.1136/bmjsem-2020-000752](https://doi.org/10.1136/bmjsem-2020-000752)
18. Manley G, Gardner AJ, Schneider KJ, et al. A systematic review of potential long-term effects of sport-related concussion. *Br J Sports Med.* 2017;51(12):969-977. doi:[10.1136/bjsports-2017-097791](https://doi.org/10.1136/bjsports-2017-097791)
19. Sinnott AM, Elbin RJ, Collins MW, Reeves VL, Holland CL, Kontos AP. Persistent vestibular-ocular impairment following concussion in adolescents. *J Sci Med Sport.* 2019;22(12):1292-1297. doi:[10.1016/j.jsams.2019.08.004](https://doi.org/10.1016/j.jsams.2019.08.004)
20. Hoge CW, McGurk D, Thomas JL, Cox AL, Engel CC, Castro CA. Mild traumatic brain injury in U.S. soldiers returning from Iraq. *N Engl J Med.* 2008;358(5):453-463. doi:[10.1056/NEJMoa072972](https://doi.org/10.1056/NEJMoa072972)
21. Alla S, Sullivan SJ, Hale L, McCrory P. Self-report scales/checklists for the measurement of concussion symptoms: a systematic review. *Br J Sports Med.* 2009;43(Suppl 1):i3-i12. doi:[10.1136/bjsm.2009.058339](https://doi.org/10.1136/bjsm.2009.058339)
22. Vander Werff KR, Rieger B. Auditory and cognitive behavioral performance deficits and symptom reporting in postconcussion syndrome following mild traumatic brain injury. *J Speech Lang Hear Res.* 2019;62:1-18. doi:[10.1044/2019_JSLHR-H-18-0281](https://doi.org/10.1044/2019_JSLHR-H-18-0281)
23. Echlin PS, Tator CH, Cusimano MD, et al. Return to play after an initial or recurrent concussion in a prospective study of physician-observed junior ice hockey concussions: implications for return to play after a concussion. *Neurosurg Focus.* 2010;29(5):E5. doi:[10.3171/2010.9.FOCUS10210](https://doi.org/10.3171/2010.9.FOCUS10210)
24. Herring S, Kibler WB, Putukian M, et al. Selected issues in sport-related concussion (SRC|mild traumatic brain injury) for the team physician: a consensus statement. *Br J Sports Med.* 2021;55(22):1251-1261. doi:[10.1136/bjsports-2021-104235](https://doi.org/10.1136/bjsports-2021-104235)
25. Heslot C, Azouvi P, Perdrieau V, Granger A, Lefevre-Dognin C, Cogne M. A systematic review of treatments of post-concussion symptoms. *J Clin Med.* 2022;11(20):6224. doi:[10.3390/jcm11206224](https://doi.org/10.3390/jcm11206224)
26. Makdissi M, Schneider KJ, Feddermann-Demont N, et al. Approach to investigation and treatment of persistent symptoms following sport-related concussion: a systematic review. *Br J Sports Med.* 2017;51(12):958-968. doi:[10.1136/bjsports-2016-097470](https://doi.org/10.1136/bjsports-2016-097470)
27. McLeod TC, Lewis JH, Whelihan K, Bacon CE. Rest and return to activity after sport-related concussion: a systematic review of the literature. *J Athl Train.* 2017;52(3):262-287. doi:[10.4085/1052-6050-51.6.06](https://doi.org/10.4085/1052-6050-51.6.06)
28. Lumba-Brown A, Yeates KO, Sarmiento K, et al. Diagnosis and management of mild traumatic brain injury in children: a systematic review. *JAMA Pediatr.* 2018;172(11):e182847. doi:[10.1001/jamapediatrics.2018.2847](https://doi.org/10.1001/jamapediatrics.2018.2847)
29. Rytter HM, Graff HJ, Henriksen HK, et al. Nonpharmacological treatment of persistent postconcussion symptoms in adults: a systematic review and meta-analysis and guideline recommendation. *JAMA Netw Open.* 2021;4(11):e2132221. doi:[10.1001/jamanetworkopen.2021.32221](https://doi.org/10.1001/jamanetworkopen.2021.32221)
30. Harmon KG, Glugston JR, Dec K, et al. American Medical Society for Sports Medicine position statement on concussion in sport. *Br J Sports Med.* 2019;53:213-225. doi:[10.1136/bjsports-2018-100338](https://doi.org/10.1136/bjsports-2018-100338)
31. Moher D, Liberati A, Tetzlaff J, Altman DG, Group P. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med.* 2009;6(7):e1000097. doi:[10.1371/journal.pmed.1000097](https://doi.org/10.1371/journal.pmed.1000097)
32. EBM tools. *Centre for Evidence-Based Medicine (CEBM)*. University of Oxford. Accessed July 7; 2022 <https://www.cebm.ox.ac.uk/resources/ebm-tools>
33. Levels of evidence. *Centre for Evidence-Based Medicine (CEBM)*. University of Oxford. Accessed May 1; 2021 <https://www.cebm.ox.ac.uk/resources/levels-of-evidence>
34. Risk of Bias Tool | Cochrane Bias. Accessed August 6 2021 <https://methods.cochrane.org/bias/risk-bias-tool>
35. Williams K, Thomson D, Seto I, et al. Standard 6: age groups for pediatric trials. *Pediatrics.* 2012;129(Suppl 3):S153-S160. doi:[10.1542/peds.2012-00551](https://doi.org/10.1542/peds.2012-00551)
36. Fitch K, Bernstein Maria SJ, Aguilar D, et al. The RAND/UCLA Appropriateness Method User's Manual. RAND Corporation. 2001 Accessed May 1, 2021. https://www.rand.org/pubs/monograph_reports/MR1269.html
37. Varner CE, McLeod S, Nahiddi N, Loughheed RE, Dear TE, Borgundvaag B. Cognitive rest and graduated return to usual activities versus usual care for mild traumatic brain injury: a randomized controlled trial of emergency department discharge instructions. *Acad Emerg Med.* 2017;24(1):75-82. doi:[10.1111/acem.13073](https://doi.org/10.1111/acem.13073)
38. Brown NJ, Mannix RC, O'Brien MJ, Gostine D, Collins MW, Meehan WP 3rd. Effect of cognitive activity level on duration of post-concussion symptoms. *Pediatrics.* 2014;133(2):e299-e304. doi:[10.1542/peds.2013-2125](https://doi.org/10.1542/peds.2013-2125)
39. Brooks BL, Low TA, Daya H, Khan S, Mikrogianakis A, Barlow KM. Test or rest? Computerized cognitive testing in the emergency department after pediatric mild traumatic brain injury does not delay symptom recovery. *J Neurotrauma.* 2016;33(23):2091-2096. doi:[10.1089/neu.2015.4301](https://doi.org/10.1089/neu.2015.4301)
40. Gibson S, Nigrovic LE, O'Brien M, Meehan WP 3rd. The effect of recommending cognitive rest on recovery from sport-related concussion. *Brain Inj.* 2013;27(7-8):839-842. doi:[10.3109/02699052.2013.775494](https://doi.org/10.3109/02699052.2013.775494)
41. Maerlender A, Rieman W, Lichtenstein J, Condoracci C. Programmed physical exertion in recovery from sports-related concussion: a randomized pilot study. *Dev Neuropsychol.* 2015;40(5):273-278. doi:[10.1080/87565641.2015.1067706](https://doi.org/10.1080/87565641.2015.1067706)
42. Grool AM, Aglipay M, Momoli F, et al. Association between early participation in physical activity following acute concussion and persistent postconcussive symptoms in children and adolescents. *JAMA.* 2016;316(23):2504-2514. doi:[10.1001/jama.2016.17396](https://doi.org/10.1001/jama.2016.17396)
43. Howell DR, Mannix RC, Quinn B, Taylor JA, Tan CO, Meehan WP 3rd. Physical activity level and symptom duration are not associated after concussion. *Am J Sports Med.* 2016;44(4):1040-1046. doi:[10.1177/0363546515625045](https://doi.org/10.1177/0363546515625045)
44. Willer BS, Haider MN, Bezherano I, et al. Comparison of rest to aerobic exercise and placebo-like treatment of acute sport-related concussion in male and female adolescents. *Arch Phys*

- Med Rehabil.* 2019;100(12):2267-2275. doi:10.1016/j.apmr.2019.07.003
45. de Kruijk JR, Leffers P, Meerhoff S, Rutten J, Twijstra A. Effectiveness of bed rest after mild traumatic brain injury: a randomised trial of no versus six days of bed rest. *J Neurol Neurosurg Psychiatry.* 2002;73(2):167-172. doi:10.1136/jnnp.73.2.167
 46. Thomas DG, Apps JN, Hoffmann RG, McCrea M, Hammke T. Benefits of strict rest after acute concussion: a randomized controlled trial. *Pediatrics.* 2015;135(2):213-223. doi:10.1542/peds.2014-0966
 47. Sufrinko AM, Kontos AP, Apps JN, et al. The effectiveness of prescribed rest depends on initial presentation after concussion. *J Pediatr.* 2017;185:167-172. doi:10.1016/j.jpeds.2017.02.072
 48. Taubman B, Rosen F, McHugh J, Grady MF, Elci OU. The timing of cognitive and physical rest and recovery in concussion. *J Child Neurol.* 2016;31(14):1555-1560. doi:10.1177/0883073816664835
 49. Root JM, Sady MD, Gai J, Vaughan CG, Madati PJ. Effect of cognitive and physical rest on persistent Postconcussive symptoms following a pediatric head injury. *J Pediatr.* 2020;227:184-190 e4. doi:10.1016/j.jpeds.2020.07.049
 50. Buckley TA, Munkasy BA, Clouse BP. Acute cognitive and physical rest may not improve concussion recovery time. *J Head Trauma Rehabil.* 2016;31(4):233-241. doi:10.1097/HTR.000000000000165
 51. Moser RS, Glatts C, Schatz P. Efficacy of immediate and delayed cognitive and physical rest for treatment of sports-related concussion. *J Pediatr.* 2012;161(5):922-926. doi:10.1016/j.jpeds.2012.04.012
 52. Moser RS, Schatz P, Glenn M, Kollias KE, Iverson GL. Examining prescribed rest as treatment for adolescents who are slow to recover from concussion. *Brain Inj.* 2015;29(1):58-63. doi:10.3109/02699052.2014.964771
 53. Leddy JJ, Haider MN, Ellis MJ, et al. Early subthreshold aerobic exercise for sport-related concussion: a randomized clinical trial. *JAMA Pediatr.* 2019;173(4):319-325. doi:10.1001/jamapediatrics.2018.4397
 54. Howell DR, Hunt DL, Aaron SE, Meehan WP 3rd, Tan CO. Influence of aerobic exercise volume on Postconcussion symptoms. *Am J Sports Med.* 2021;49(7):1912-1920. doi:10.1177/03635465211005761
 55. Leddy JJ, Master CL, Mannix R, et al. Early targeted heart rate aerobic exercise versus placebo stretching for sport-related concussion in adolescents: a randomised controlled trial. *Lancet Child Adolesc Health.* 2021;5(11):792-799. doi:10.1016/S2352-4642(21)00267-4
 56. Varner CE, Thompson C, de Wit K, Borgundvaag B, Houston R, McLeod S. A randomized trial comparing prescribed light exercise to standard management for emergency department patients with acute mild traumatic brain injury. *Acad Emerg Med.* 2021;28(5):493-501. doi:10.1111/acem.14215
 57. Howell DR, Wingerson MJ, Kirkwood MW, Grubenhoff JA, Wilson JC. Early aerobic exercise among adolescents at moderate/high risk for persistent post-concussion symptoms: a pilot randomized clinical trial. *Phys Ther Sport.* 2022;55:196-204. doi:10.1016/j.ptsp.2022.04.010
 58. Worts PR, Mason JR, Burkhart SO, Sanchez-Gonzalez MA, Kim JS. The acute, systemic effects of aerobic exercise in recently concussed adolescent student-athletes: preliminary findings. *Eur J Appl Physiol.* 2022;122(6):1441-1457. doi:10.1007/s00421-022-04932-4
 59. Howell DR, Brilliant AN, Oldham JR, Berkstresser B, Wang F, Meehan WP 3rd. Exercise in the first week following concussion among collegiate athletes: preliminary findings. *J Sci Med Sport.* 2020;23(2):112-117. doi:10.1016/j.jsams.2019.08.294
 60. Leddy JJ, Haider MN, Hinds AL, Darling S, Willer BS. A preliminary study of the effect of early aerobic exercise treatment for sport-related concussion in males. *Clin J Sport Med.* 2019;29(5):353-360. doi:10.1097/JSM.0000000000000663
 61. Seehusen CN, Wilson JC, Walker GA, Reinking SE, Howell DR. More physical activity after concussion is associated with faster return to play among adolescents. *Int J Environ Res Public Health.* 2021;18(14):7373. doi:10.3390/ijerph18147373
 62. Yang J, Yeates KO, Shi J, et al. Association of Self-Paced Physical and Cognitive Activities across the first week Postconcussion with symptom resolution in youth. *J Head Trauma Rehabil.* 2021;36(2):E71-E78. doi:10.1097/HTR.0000000000000642
 63. Remigio-Baker RA, Bailie JM, Gregory E, et al. Activity level during acute concussion may predict symptom recovery within an active duty military population. *J Head Trauma Rehabil.* 2020;35(2):92-103. doi:10.1097/HTR.0000000000000498
 64. Buckley TA, Munkasy BA, Evans KM, Clouse B. Acute physical and mental activity influence on concussion recovery. *Med Sci Sports Exerc.* 2022;54(2):307-312. doi:10.1249/MSS.0000000000002787
 65. Majerske CW, Mihalik JP, Ren D, et al. Concussion in sports: postconcussive activity levels, symptoms, and neurocognitive performance. *J Athl Train.* 2008;43(3):265-274. doi:10.4085/1062-6050.43.3.265
 66. Lawrence DW, Richards D, Comper P, Hutchison MG. Earlier time to aerobic exercise is associated with faster recovery following acute sport concussion. *PLoS One.* 2018;13(4):e0196062. doi:10.1371/journal.pone.0196062
 67. Popovich M, Almeida A, Freeman J, et al. Use of supervised exercise during recovery following sports-related concussion. *Clin J Sport Med.* 2021;31(2):127-132. doi:10.1097/JSM.0000000000000721
 68. Stumph J, Young J, Singichetti B, et al. Effect of exercise recommendation on adolescents with concussion. *J Child Neurol.* 2020;35(2):95-101. doi:10.1177/0883073819877790
 69. Chan C, Iverson GL, Purtzki J, et al. Safety of active rehabilitation for persistent symptoms after pediatric sport-related concussion: a randomized controlled trial. *Arch Phys Med Rehabil.* 2018;99(2):242-249. doi:10.1016/j.apmr.2017.09.108
 70. Kurowski BG, Hugentobler J, Quatman-Yates C, et al. Aerobic exercise for adolescents with prolonged symptoms after mild traumatic brain injury: an exploratory randomized clinical trial. *J Head Trauma Rehabil.* 2017;32(2):79-89. doi:10.1097/HTR.0000000000000238
 71. Snyder AR, Greif SM, Clugston JR, et al. The effect of aerobic exercise on concussion recovery: a pilot clinical trial. *J Int Neuropsychol Soc.* 2021;27(8):790-804. doi:10.1017/S1355617721000886
 72. Leddy JJ, Cox JL, Baker JG, et al. Exercise treatment for post-concussion syndrome: a pilot study of changes in functional magnetic resonance imaging activation, physiology, and symptoms. *J Head Trauma Rehabil.* 2013;28(4):241-249. doi:10.1097/HTR.0b013e31826da964
 73. Larson-Dupuis C, Leveille E, Desjardins M, et al. Subtle long-term cognitive effects of a single mild traumatic brain injury and the impact of a three-month aerobic exercise intervention. *J Sports Med Phys Fitness.* 2021;61(1):87-95. doi:10.23736/S0022-4707.20.10918-6
 74. Leddy JJ, Kozlowski K, Donnelly JP, Pendergast DR, Epstein LH, Willer B. A preliminary study of subsymptom threshold exercise training for refractory post-concussion syndrome. *Clin J Sport Med.* 2010;20(1):21-27. doi:10.1097/JSM.0b013e3181c6c22c
 75. Gagnon I, Grilli L, Friedman D, Iverson GL. A pilot study of active rehabilitation for adolescents who are slow to recover from sport-related concussion. *Scand J Med Sci Sports.* 2016;26(3):299-306. doi:10.1111/sms.12441
 76. Moore BM, Adams JT, Barakatt E. Outcomes following a vestibular rehabilitation and aerobic training program to address persistent post-concussion symptoms. *J Allied Health Winter.* 2016;45(4):e59-e68.
 77. Coslick AM, Chin KE, Kalb LG, Slomine BS, Suskauer SJ. Participation in physical activity at time of presentation to a specialty concussion clinic is associated with shorter time to recovery. *PM R.* 2020;12(12):1195-1204. doi:10.1002/pmrj.12443

78. Cordingley D, Girardin R, Reimer K, et al. Graded aerobic treadmill testing in pediatric sports-related concussion: safety, clinical use, and patient outcomes. *J Neurosurg Pediatr*. 2016;25(6):693-702. doi:10.3171/2016.5.PEDS16139
79. Chrisman SPD, Whitlock KB, Somers E, et al. Pilot study of the Sub-Symptom Threshold Exercise Program (SSTEP) for persistent concussion symptoms in youth. *NeuroRehabilitation*. 2017;40(4):493-499. doi:10.3233/NRE-161436
80. Dobney DM, Grilli L, Kocilowicz H, et al. Evaluation of an active rehabilitation program for concussion management in children and adolescents. *Brain Inj*. 2017;31(13-14):1753-1759. doi:10.1080/02699052.2017.1346294
81. Grabowski P, Wilson J, Walker A, Enz D, Wang S. Multimodal impairment-based physical therapy for the treatment of patients with post-concussion syndrome: a retrospective analysis on safety and feasibility. *Phys Ther Sport*. 2017;23:22-30. doi:10.1016/j.ptsp.2016.06.001
82. Van Vleet TM, Chen A, Vernon A, Novakovic-Agopian T, D'Esposito MT. Tonic and phasic alertness training: a novel treatment for executive control dysfunction following mild traumatic brain injury. *Neurocase*. 2015;21(4):489-498. doi:10.1080/13554794.2014.928329
83. O'Connor MK, Mueller L, Kwon E, et al. Enhanced vocational rehabilitation for veterans with mild traumatic brain injury and mental illness: pilot study. *J Rehabil Res Dev*. 2016;53(3):307-320. doi:10.1682/JRRD.2014.10.0231
84. Vas A, Chapman S, Aslan S, et al. Reasoning training in veteran and civilian traumatic brain injury with persistent mild impairment. *Neuropsychol Rehabil*. 2016;26(4):502-531. doi:10.1080/09602011.2015.1044013
85. Cooper DB, Bowles AO, Kennedy JE, et al. Cognitive rehabilitation for military service members with mild traumatic brain injury: a randomized clinical trial. *J Head Trauma Rehabil*. 2017;32(3):E1-E15. doi:10.1097/HTR.0000000000000254
86. Han K, Chapman SB, Krawczyk DC. Neuroplasticity of cognitive control networks following cognitive training for chronic traumatic brain injury. *Neuroimage Clin*. 2018;18:262-278. doi:10.1016/j.nicl.2018.01.030
87. Mahncke HW, DeGutis J, Levin H, et al. A randomized clinical trial of plasticity-based cognitive training in mild traumatic brain injury. *Brain*. 2021;144(7):1994-2008. doi:10.1093/brain/awab202
88. Teel E, Brossard-Racine M, Corbin-Berrigan LA, Gagnon I. Perceptual cognitive training does not improve clinical outcomes at 4 and 12 weeks following concussion in children and adolescents: a randomized controlled trial. *J Head Trauma Rehabil*. 2021;36(2):E97-E107. doi:10.1097/HTR.0000000000000633
89. Thiagarajan P, Ciuffreda KJ. Effect of oculomotor rehabilitation on vergence responsivity in mild traumatic brain injury. *J Rehabil Res Dev*. 2013;50(9):1223-1240. doi:10.1682/JRRD.2012.12.0235
90. Thiagarajan P, Ciuffreda KJ. Effect of oculomotor rehabilitation on accommodative responsivity in mild traumatic brain injury. *J Rehabil Res Dev*. 2014;51(2):175-191. doi:10.1682/JRRD.2013.01.0027
91. Thiagarajan P, Ciuffreda KJ, Capo-Aponte JE, Ludlam DP, Kapoor N. Oculomotor neurorehabilitation for reading in mild traumatic brain injury (mTBI): an integrative approach. *NeuroRehabilitation*. 2014;34(1):129-146. doi:10.3233/NRE-131025
92. Thiagarajan P, Ciuffreda KJ. Versional eye tracking in mild traumatic brain injury (mTBI): effects of oculomotor training (OMT). *Brain Inj*. 2014;28(7):930-943. doi:10.3109/02699052.2014.888761
93. Ciuffreda KJ, Rutner D, Kapoor N, Suchoff IB, Craig S, Han ME. Vision therapy for oculomotor dysfunctions in acquired brain injury: a retrospective analysis. *Optometry*. 2008;79(1):18-22. doi:10.1016/j.optm.2007.10.004
94. Yadav NK, Thiagarajan P, Ciuffreda KJ. Effect of oculomotor vision rehabilitation on the visual-evoked potential and visual attention in mild traumatic brain injury. *Brain Inj*. 2014;28(7):922-929. doi:10.3109/02699052.2014.887227
95. Rosner MS, Feinberg DL, Doble JE, Rosner AJ. Treatment of vertical heterophoria ameliorates persistent post-concussive symptoms: a retrospective analysis utilizing a multi-faceted assessment battery. *Brain Inj*. 2016;30(3):311-317. doi:10.3109/02699052.2015.1113564
96. Schneider KJ, Meeuwisse WH, Nettel-Aguirre A, et al. Cervicovestibular rehabilitation in sport-related concussion: a randomised controlled trial. *Br J Sports Med*. 2014;48(17):1294-1298. doi:10.1136/bjsports-2013-093267
97. Kontos AP, Eagle SR, Mucha A, et al. A randomized controlled trial of precision vestibular rehabilitation in adolescents following concussion: preliminary findings. *J Pediatr*. 2021;239:193-199. doi:10.1016/j.jpeds.2021.08.032
98. Alsalaheen B, Carender W, Grzesiak M, et al. Changes in vestibular/ocular-motor screen scores in adolescents treated with vestibular therapy after concussion. *Pediatr Phys Ther*. 2020;32(4):331-337. doi:10.1097/PEP.0000000000000729
99. Ahluwalia R, Miller S, Dawoud FM, et al. A pilot study evaluating the timing of vestibular therapy after sport-related concussion: is earlier better? *Sports Health*. 2021;13(6):573-579. doi:10.1177/1941738121998687
100. Alsalaheen BA, Mucha A, Morris LO, et al. Vestibular rehabilitation for dizziness and balance disorders after concussion. *J Neurol Phys Ther*. 2010;34(2):87-93. doi:10.1097/NPT.0b013e3181d0e568
101. Kleffelgaard I, Soberg HL, Bruusgaard KA, Tamber AL, Langhammer B. Vestibular rehabilitation after traumatic brain injury: case series. *Phys Ther*. 2016;96(6):839-849. doi:10.2522/ptj.20150095
102. Prangley A, Aggerholm M, Cinelli M. Improvements in balance control in individuals with PCS detected following vestibular training: a case study. *Gait Posture*. 2017;58:229-231. doi:10.1016/j.gaitpost.2017.08.006
103. Elgmark Andersson E, Emanuelson I, Bjorklund R, Stalhammar DA. Mild traumatic brain injuries: the impact of early intervention on late sequelae. A randomized controlled trial. *Acta Neurochir (Wien)*. 2007;149(2):151-159; discussion 160. doi:10.1007/s00701-006-1082-0
104. Dharm-Datta S, Gough MR, Porter PJ, et al. Successful outcomes following neurorehabilitation in military traumatic brain injury patients in the United Kingdom. *J Trauma Acute Care Surg*. 2015;79(4 Suppl 2):S197-S203. doi:10.1097/TA.0000000000000721
105. Reneker JC, Hassen A, Phillips RS, Moughiman MC, Donaldson M, Moughiman J. Feasibility of early physical therapy for dizziness after a sports-related concussion: a randomized clinical trial. *Scand J Med Sci Sports*. 2017;27(12):2009-2018. doi:10.1111/sms.12827
106. McCarty CA, Zatzick DF, Marcynyszyn LA, et al. Effect of collaborative care on persistent postconcussive symptoms in adolescents: a randomized clinical trial. *JAMA Netw Open*. 2021;4(2):e210207. doi:10.1001/jamanetworkopen.2021.0207
107. Langevin P, Fremont P, Fait P, Dube MO, Bertrand-Charette M, Roy JS. Cervicovestibular rehabilitation in adults with mild traumatic brain injury: a randomized clinical trial. *J Neurotrauma*. 2022;39(7-8):487-496. doi:10.1089/neu.2021.0508
108. Gauvin-Lepage J, Friedman D, Grilli L, et al. Effectiveness of an exercise-based active rehabilitation intervention for youth who are slow to recover after concussion. *Clin J Sport Med*. 2020;30(5):423-432. doi:10.1097/JSM.0000000000000634
109. Hammerle M, Swan AA, Nelson JT, Treleaven JM. Retrospective review: effectiveness of cervical proprioception retraining for dizziness after mild traumatic brain injury in a military population with abnormal cervical proprioception. *J Manipulative Physiol Ther*. 2019;42(6):399-406. doi:10.1016/j.jmpt.2018.12.002
110. Thastum MM, Rask CU, Naess-Schmidt ET, et al. Design of an early intervention for persistent post-concussion symptoms in adolescents and young adults: a feasibility study. *NeuroRehabilitation*. 2018;43(2):155-167. doi:10.3233/NRE-172391

111. Ledoux AA, Barrowman N, Bijelic V, et al. Is early activity resumption after paediatric concussion safe and does it reduce symptom burden at 2 weeks post injury? The pediatric concussion assessment of rest and exertion (PedCARE) multicentre randomised clinical trial. *Br J Sports Med*. 2022;56(5):271-278. doi:10.1136/bjsports-2021-105030
112. Wiebe DJ, Bretzin AC, D'Alonzo BA, the Ivy League-Big Ten Epidemiology of Concussion Study I. Progression through return-to-sport and return-to-academics guidelines for concussion management and recovery in collegiate student athletes: findings from the ivy league-big ten epidemiology of concussion study. *Br J Sports Med*. 2022;56(14):801-811. doi:10.1136/bjsports-2021-104451
113. Philipson EB, Gause E, Conrick KM, et al. Concussion symptoms and temporary accommodations using a student-centered return to learn care plan. *NeuroRehabilitation*. 2021;49(4):655-662. doi:10.3233/NRE-210182
114. Kelleher E, Taylor-Linzey E, Ferrigno L, Bryson J, Kaminski S. A community return-to-play mTBI clinic: results of a pilot program and survey of high school athletes. *J Pediatr Surg*. 2014;49(2):341-344. doi:10.1016/j.jpedsurg.2013.10.016
115. Darling SR, Leddy JJ, Baker JG, et al. Evaluation of the Zurich guidelines and exercise testing for return to play in adolescents following concussion. *Clin J Sport Med*. 2014;24(2):128-133. doi:10.1097/JSM.0000000000000026
116. Meier TB, Brummel BJ, Singh R, Nerio CJ, Polanski DW, Bellgowan PS. The underreporting of self-reported symptoms following sports-related concussion. *J Sci Med Sport*. 2015;18(5):507-511. doi:10.1016/j.jsams.2014.07.008
117. DiFazio M, Silverberg ND, Kirkwood MW, Bernier R, Iverson GL. Prolonged activity restriction after concussion: are we worsening outcomes? *Clin Pediatr (Phila)*. 2016;55(5):443-451. doi:10.1177/0009922815589914
118. PubMed. Accessed October 13, 2022. <https://pubmed.ncbi.nlm.nih.gov/>.
119. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: De Luigi AJ, Bell KR, Bramhall JP, et al. Consensus statement: An evidence-based review of exercise, rehabilitation, rest, and return to activity protocols for the treatment of concussion and mild traumatic brain injury. *PM&R*. 2023;15(12):1605-1642. doi:10.1002/pmrj.13070

CME Question

Based on this review, in adolescents with acute concussion/mTBI, subsymptom threshold aerobic exercise is an appropriate intervention and

- A. Contributes to symptomatic improvement
- B. Determines clearance for return to activity
- C. Provides similar benefits when compared to other age groups
- D. Is comparable to other therapeutic interventions in benefits

Answer online at <https://onlinelearning.aapmr.org/>

This journal-based CME activity is designated for 1.0 AMA PRA Category 1 Credit and can be completed online at <https://onlinelearning.aapmr.org/>. This activity is FREE to AAPM&R members and available to nonmembers for a nominal fee. CME is available for 3 years after publication date. For assistance with claiming CME for this activity, please contact (847) 737-6000.

All financial disclosures and CME information related to this article can be found on the Online Learning Portal (<https://onlinelearning.aapmr.org/>) prior to accessing the activity.